Diagnosing Mitigation Need: Benchmark Results for Controlled Shortages and Signal-Matched Interventions in Histopathology Patch Classification

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Abstract

A class can be underserved for several distinct reasons, and the count that marks it as rare does not say which. This report evaluates the controlled patch-classification benchmark specified in the companion protocol (Queisler, 2026a) on BRACS, a seven-class breast-lesion task represented by Virchow2 features. The design separates two axes that ordinary imbalance confounds: nominal patch support and the number of patients contributing those patches. Nominal deprivation produced material balanced-accuracy losses in eight of nine comparisons, with deficits of 1.5–13.8 percentage points; tail-group likelihood also worsened in eight comparisons, by up to 1.744 nats (natural-log units of predictive loss), although a single validation-fitted temperature removes most of that likelihood deficit and leaves three of the eight above the materiality minimum. Reducing patient breadth while holding the patch allocation fixed produced balanced-accuracy deficits of 2.6–3.0 percentage points across all three class assignments. Prevalence-weighted and class-balanced cross-entropy were the best methods for nominal damage. The largest probability-quality recovery estimate, 1.23 for class-balanced cross-entropy, means that the method surpassed the balanced reference by 23% of the original deficit after closing it; it does not imply that more than 100% of the lost probability mass was recovered. By contrast, no tested method repaired the damage caused by reduced patient breadth, and signal-matched weighting differed from unmatched weighting by only 0.06 percentage points of balanced accuracy. These results motivate mitigation that responds to dependence among patches from the same patient rather than to per-class counts alone.

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1 Introduction

This report presents the results of the controlled patch-classification benchmark whose design, endpoints, materiality criteria, and inference procedure are specified in the companion protocol (Queisler, 2026a). The benchmark exists to inform the design of a new mitigation method, not to rank the existing ones. Its premise is that an underserved class may lack nominal patch support, independent evidence from patients or slides, learnable signal, or representational diversity, and that these shortages are not interchangeable. Existing methods are therefore treated as instruments: each reads one signal, and the contrast among them measures whether reading the right signal matters. The report is consequently read forward into Section 5, where the outcomes are converted into requirements a method would have to satisfy. Dataset provenance is documented separately (Queisler, 2026b), as is the signal taxonomy that locates each method within it (Queisler, 2026c).

The protocol poses three questions, restated here so that the results can be read without the companion document at hand:

RQ1. Which shortage does controlled allocation create, and what does it damage? Under otherwise fixed conditions, how do controlled changes in training allocation affect tail-class discrimination and calibration, and which signal profile does each condition realize?

RQ2. Does signal-matched mitigation recover more of the damage? Which methods recover performance lost specifically to training imbalance, and does recovery improve when a method reads the shortage identified in RQ1?

RQ3. Which realized signal shortages explain variation in damage and mitigation headroom? Across controlled conditions, are allocation severity, independent-evidence shortage, support–difficulty alignment, and diversity shortage associated with damage magnitude and subsequent recovery?

Sections 2 and 3 answer the first two. Section 4 specifies the association models for the third. Statistical tests are restricted to the five weighting methods and to comparisons meeting the prespecified materiality and precision criteria; estimates for the wider method roster remain exploratory.

2 Allocation-Induced Damage

Four prerequisites must hold before the outcome measures are interpreted:

1. the moderate and severe allocations must reach their target ranges;
2. the concentrated and spread allocations must differ in patient breadth while retaining identical patch counts;
3. the pooled analysis must contain enough contributing patients for stable interval estimates; and
4. every planned method, initialization seed, and data partition must be present.

A failed allocation would invalidate the intended comparison, insufficient patient support would limit the corresponding estimates to summaries, and incomplete results would not be aggregated. Appendix A provides the full checks, tuning selections, and protocol deviations. The companion protocol describes the four deprived conditions and their two reference allocations (Queisler, 2026a).

All four prerequisites were met. Moderate and severe allocations achieved $\rho = 10.00$ and 100.38, respectively, within their target ranges. The spread allocation retained exactly 16,796 patches at $\rho = 1.000$ while increasing the contributing-patient count from 85 to the full eligible inventory of 103. Every class had sufficient patient support in the pooled analysis. All three partitions contained five initialization seeds for every method block and the complete set of 378 comparisons. The analysis therefore contains 12 comparison units: three class assignments crossed with four deprived conditions.

2.1 Realized signal profiles

Allocation severity is the manipulated variable, but it is not the explanatory one. Each controlled condition is characterized instead by the signal profile it realizes: achieved imbalance ratio ρ and normalized-entropy deficit $1 - H_{\text{norm}}$, nominal patch-support shortage, independent patient-support shortage, support–difficulty alignment, and representational-diversity shortage. These quantities are measured from the realized training sets rather than inferred from the target ratio. Each condition is compared with the reference that isolates its axis: concentrated balanced for the concentrated moderate and severe conditions, and balanced spread for concentrated balanced and severe spread.

Assignment	ρ	$1 - H_{\text{norm}}$	S_{nom}	S_{ind}	S_{div}	A	Dominant
Concentrated balanced vs balanced spread							
Native	1.00	0.000	-1.47	1.73	-1.32	0.20	Independent support
Aligned	1.00	0.000	-1.47	1.73	-1.32	0.20	Independent support
Reversed	1.00	0.000	-1.47	1.73	-1.31	0.20	Independent support
Concentrated moderate vs concentrated balanced							
Native	10.00	0.128	-0.35	-0.58	-0.79	0.36	Nominal support
Aligned	10.00	0.128	-0.17	-0.58	-0.35	-0.92	Difficulty
Reversed	10.00	0.128	-0.39	-0.58	-0.65	0.92	Nominal support
Concentrated severe vs concentrated balanced							
Native	100.38	0.353	0.76	-0.58	0.80	0.36	Ambiguous
Aligned	100.38	0.353	1.26	-0.58	1.28	-0.92	Difficulty
Reversed	100.38	0.353	0.65	-0.58	0.79	0.92	Ambiguous
Severe spread vs balanced spread							
Native	100.38	0.353	0.76	-0.58	1.00	0.36	Ambiguous
Aligned	100.38	0.353	1.26	-0.58	0.88	-0.92	Difficulty
Reversed	100.38	0.353	0.65	-0.58	0.99	0.92	Diversity

Table 1: BRACS signal profiles for the four deprived conditions. Each block names the deprived allocation and its reference; rows give the three class assignments. Shortage scores and alignment are standardized across the 12 comparison units. A dominant shortage is marked ambiguous when the two largest scores differ by less than 0.25 standard deviations.

The independent-support comparison holds 16,796 patches fixed but draws them from fewer patients. Across partitions, the affected classes retain about 51–66% of the patient support available in the spread reference, corresponding to a mean log-shortage of 0.409–0.673 nats. This is the largest patient-breadth contrast compatible with the contribution limits.

The three concentrated-balanced rows reuse one training allocation and differ only in the class assignment used for comparison; they are correlated views, not independent replications. Three severe comparisons are ambiguous because their two largest shortage scores are nearly tied: native severe, reversed severe, and native severe spread.

2.2 Discrimination deficit

The discrimination deficit D_{BA} is the balanced accuracy lost relative to the appropriate reference, reported in percentage points. The native assignment follows clinical class order; the difficulty-aligned assignment gives the fewest patches to the hardest classes, whereas the reversed assignment gives those classes the most patches. The 1.44-point minimum is $\max(1, 2\sigma_{\mathrm{seed}})$: the larger of the one-point pilot materiality floor and twice the balanced-condition standard deviation across confirmation seeds. Recovery is analysed when the deficit reaches this minimum and its paired 95% interval excludes zero.

Moderate imbalance produces deficits of 1.50–8.17 percentage points, concentrated severe imbalance 4.30–13.23, and severe spread 4.15–13.77. Concentrating the balanced allocation on fewer patients costs 2.62–3.03 percentage points despite identical patch counts. The only exception is difficulty-aligned moderate: its 1.50-point estimate clears the magnitude threshold, but its interval includes zero. Full estimates and intervals appear in Appendix Table 17.

2.3 Probability-quality deficit

Probability quality is measured by tail-group macro negative log likelihood. One *nat* is one natural-log unit of predictive loss, and lower values are better. The 0.148-nat minimum is likewise derived from balanced-condition variation, as twice the confirmation-seed standard deviation. Recovery is analysed when deprivation reaches this minimum and its paired 95% interval excludes zero.

Moderate imbalance increases loss by 0.186–0.754 nats, concentrated severe imbalance by 0.688–1.744, and severe spread by 0.205–1.129. Native moderate is the exception because its interval includes zero. The independent-support comparisons increase loss by 0.134–0.140 nats, just below the magnitude threshold, so they do not enter probability-quality recovery analysis. Full estimates, intervals, and secondary calibration measures appear in Appendix Table 18.

2.4 Where the damage falls

Aggregate deficits do not show which support tier absorbs the loss. Table 2 therefore retains only recall for the two difficulty-based assignments; full per-class endpoints appear in Appendix D. Tier labels in the balanced spread rows reflect class ordering rather than unequal support because that allocation is uniform.

	Aligned recall (%)			Reversed recall (%)		
Condition	Head	Body	Tail	Head	Body	Tail
Balanced spread	67.0	32.5	40.9	40.4	31.2	66.9
Moderate	76.4	37.6	18.3	46.1	19.7	45.9
Severe	74.7	41.0	11.8	48.5	15.3	35.5
Severe spread	73.8	40.8	13.2	51.2	13.3	31.9

Table 2: Tier recall (%) under aligned and reversed assignments.

Under the aligned assignment, tail recall falls from 40.8% under balanced spread to 18.3% at moderate and 11.8% at severe imbalance, while head recall remains above 73%. Under the reversed assignment, the body tier is weakest: recall falls from 31.0% under balanced spread to 15.3% at severe and 13.3% at severe spread. Difficulty therefore changes where damage appears; support rank alone does not identify the affected tier.

2.5 Natural anchor

The natural condition trains on all 217,399 eligible patches at the dataset's observed prevalence. It is not size-matched and enters no estimand; it only places the controlled results beside the training distribution encountered without intervention.

Condition	Training patches	Balanced accuracy (%)
Natural	217,399	49.8
Concentrated balanced	16,796	49.2
Balanced spread	16,796	50.4–50.9

Table 3: Natural and balanced cross-entropy anchors.

Despite using nearly thirteen times as many patches, the natural condition reaches 49.8% balanced accuracy, compared with 49.2% for concentrated balanced and 50.4–50.9% across the balanced spread assignments. Additional patches at natural prevalence therefore buy less discrimination than distributing the controlled budget across more patients.

3 Mitigation Recovery

Recovery is defined as $R_M = [M(\text{method}) - M(\text{deprived CE})]/D_M$ and is reported as a percentage of the established deficit. A recovery of 0% means no improvement over deprived cross-entropy, 100% reaches the balanced reference, a negative value means further deterioration, and a value above 100% surpasses the reference. Comparisons that do not meet the materiality and precision criteria in Sections 2.2 and 2.3 are omitted because their recovery denominator is not sufficiently stable.

3.1 Which weighting signals recover discrimination

The confirmatory family contains five weighted cross-entropy methods that differ only in the signal used to set their class weights: prevalence, nominal support, independent support, pilot difficulty, or semantic diversity. Each is compared with deprived cross-entropy using a paired permutation test. A result *survives Holm adjustment* when its adjusted p-value remains below 0.05 after accounting for all confirmatory comparisons, reducing the chance that a result appears significant only because many methods and conditions were tested.

Condition group	Units	Prevalence	Nominal support	Other signals	Survive Holm
Independent support	3	0.3%	0.4–0.5%	−6.5–2.5%	0/15
Nominal, moderate	2	81.5–84.0%	83.3–89.2%	−7.8–19.7%	4/10
Nominal, severe	3	65.4–79.4%	42.6–73.6%	−27.3–16.4%	2/15
Nominal, severe spread	3	46.6–87.1%	45.5–83.1%	−14.5–11.1%	4/15

Table 4: Balanced-accuracy recovery by shortage and weighting signal. Ranges are percentages of the original deficit; the final column counts results that remain significant after Holm adjustment.

Only prevalence-weighted and class-balanced cross-entropy consistently repair nominal-support damage. They recover between 42.6% and 89.2% across the eight eligible nominal conditions, and ten of their sixteen comparisons survive Holm adjustment. None of the 24 comparisons for independent-support, difficulty, or diversity weighting survives. Under independent-support deprivation, all five methods remain close to no recovery and none survives adjustment. Appendix Table 20 provides the condition-level estimates and intervals.

3.2 Matched against unmatched signals

A method is *matched* when its weighting signal corresponds to the dominant shortage identified for that condition in Table 1; for example, independent-support weighting is matched to a condition that reduces patient breadth. The unmatched arm contains the other weighting signals. Conditions without one clear dominant shortage are excluded from this contrast. A positive matched-minus-unmatched value therefore favours the signal selected by the condition profile.

Dominant shortage	Units	Matched minus unmatched balanced accuracy (percentage points)	Holm result
Independent support	3	0.06 [−0.13, 0.22]	No difference
Nominal support	2	2.34 [0.96, 3.75]; 7.08 [5.25, 9.15]	Both favour matched
Difficulty	2	−0.42 [−1.63, 0.86]; −1.27 [−2.42, 0.05]	No difference
Diversity	1	−5.29 [−6.48, −4.18]	Favours unmatched

Table 5: Matched-minus-unmatched contrasts for balanced accuracy. Intervals are paired 95% bootstrap intervals.

For the independent-support conditions, matched weighting changes balanced accuracy by only 0.06 percentage points, with an interval from −0.13 to 0.22 percentage points. For example, this is a change from 49.30% to 49.36%. Matching provides a benefit only in the two nominal-support conditions. When difficulty or diversity is dominant, the matched method is no better and can be worse because the unmatched arm contains the effective prevalence-based methods.

Both methods use an effective-number formula. Its parameter β controls how strongly class weights differ. The *matched-β diagnostic* reruns independent-support weighting with the value selected for class-balanced cross-entropy. Holding β constant isolates the use of patient counts rather than patch counts from the effect of selecting different values. This changes balanced accuracy by at most 1.3 percentage points across the 12 conditions and still leaves a gap of up to 11.2 percentage points to class-balanced cross-entropy. Weakly tuned independent-support weights therefore do not explain the failure. Full matched contrasts and matched-β results appear in Appendix Tables 22 and 23.

3.3 Recovery of probability quality

A method can restore balanced accuracy while making its probabilities worse. The prespecified recovery endpoint is tail-group macro negative log likelihood computed from raw test probabilities; temperature scaling never enters the gate or the confirmatory tests. It is analysed for the eight conditions with a sufficiently large and precise raw-score deficit.

Methods and conditions	Units	Recovery	Holm result
Prevalence or nominal support; all except native severe spread	7	53.9–122.7%	14/14 improvements
Prevalence or nominal support; native severe spread	1	−44.2 to −40.7%	Neither differs from zero
Independent support, difficulty, or diversity	8	−405.8–−27.4%	24/24 deteriorations

Table 6: Recovery of tail-group probability quality. Recovery is the percentage of the original negative-log-likelihood deficit.

Prevalence-weighted and class-balanced cross-entropy improve raw probability quality in seven of eight conditions, and all 14 comparisons survive Holm adjustment. The exception is native severe spread, where neither method improves on deprived cross-entropy despite recovering more than 80% of its balanced-accuracy deficit. Independent-support, difficulty, and diversity weighting worsen raw probability quality in every condition, and all 24 deteriorations survive adjustment. Exact intervals and adjusted p-values appear in Appendix Table 25.

Some of that damage may be easy to undo. *Temperature scaling* divides a model's output scores by one number fitted on validation data before they are turned into probabilities. It cannot change which class is predicted, so discrimination is untouched; it corrects only systematic over- or under-confidence. Fitting one temperature per condition lowers expected calibration error across the eight deprived cross-entropy conditions from 0.143–0.313 to 0.048–0.079, and overall negative log likelihood from 1.203–2.582 to 1.047–1.271. Whether the tail group is repaired along with the aggregate is a separate question, answered by re-estimating the tail-group contrast on scaled probabilities as Appendix E describes. A change counts below only when its paired 95% interval excludes zero.

Three findings follow, and they do not point the same way. First, most of the allocation-induced deficit is over-confidence rather than lost probability quality. After scaling, the eight deficits span -0.189 to 0.280 nats; only difficulty-reversed severe, difficulty-reversed severe spread, and difficulty-aligned severe spread still clear the 0.148-nat minimum, and difficulty-aligned moderate, difficulty-aligned severe, and native severe change sign, meaning deprived cross-entropy ends up better calibrated on the tail than its own balanced reference. Second, the repair credited to prevalence-based weighting largely goes with it: 14 of its 16 raw improvements qualify, against one after scaling, and four cells turn into deteriorations. Third, the harm done by the other three signals stays. 21 of 24 raw deteriorations qualify, and 21 of 24 scaled ones do; scaling compresses them from as much as 2.604 nats to at most 0.501 without removing them. Post-hoc calibration therefore undoes much of what the allocation did, and much of what prevalence weighting was credited with repairing, but not the damage the remaining signals cause.

3.4 Exploratory methods beyond weighting

The wider roster varies the training objective, sampling distribution, decision rule, or classifier head. Its estimates are exploratory because methods no longer differ by signal alone; several also reduce to the same balanced-sampling mechanism and should not be read as independent confirmations.

Endpoint and shortage	Units	Most informative result
Balanced accuracy, nominal support	8	Best method recovers 79.2–90.8% in seven units; 55.4% in aligned severe spread
Probability quality, nominal support	8	Best method recovers 82.2–135.6% in every unit
Balanced accuracy, independent support	3	No positive recovery with an interval excluding zero; OKO, focal loss, and cRT worsen performance

Table 7: Most informative recovery results from the exploratory method roster.

The roster shows that most nominal-support deficits are repairable by several mechanisms. The aligned severe-spread condition is harder, with a best balanced-accuracy recovery of 55.4%. Independent-support damage is different: none of the nine exploratory methods has positive recovery with an interval excluding zero, while set learning, focal loss, and classifier retraining reliably worsen balanced accuracy. Thus no tested signal or mechanism repairs the loss caused by reduced patient breadth. Full method-level estimates appear in Appendix Table 27.

4 Explaining Heterogeneity

Damage magnitude and recovery headroom vary across conditions, and RQ3 asks which realized shortages track that variation. Two association models are specified, one for signed balanced-accuracy damage over the cross-entropy cells and one for recovery over the gate-passing method cells. Both use the same linear predictor over standardized quantities,

$$\mu_j = \alpha + u_{g[j]} + \beta_\rho \log \rho_j + \beta_{\text{ind}} S_{\text{ind},j} + \beta_A A_j + \beta_{\text{div}} S_{\text{div},j}, \tag{1}$$

where ρ_j is achieved severity, $S_{\text{ind},j}$ the independent-support shortage, A_j the support–difficulty alignment, $S_{\text{div},j}$ the diversity shortage, and $u_{g[j]}$ a random intercept over dataset–target groups that absorbs level differences between datasets. Observation errors come from the bootstrap distributions of the corresponding endpoints.

The BRACS cells are complete, but the combined models await the planned TCGA-UT benchmark because the between-dataset variance is not identifiable from one dataset–target group. The recorded BRACS cells allow the combined fit to be assembled without repeating its bootstrap. Binary targets remain excluded because support–difficulty alignment over two classes is always ± 1 and would confound the pooled coefficient.

Placeholder: TCGA-UT estimates and the combined RQ3 fit will be added here after that benchmark is complete.

What can be said without the models is what Section 2.2 establishes descriptively. Damage rises monotonically with achieved severity under all three assignments; it is roughly three times larger under the difficulty-reversed assignment than under the aligned one at matched severity; and the independent-support axis produces damage at a fifth to a quarter of the nominal dose, which is the single observation in this report that a nominal-only account of shortage cannot absorb.

5 Discussion

The preceding sections establish what the controlled allocations damaged and how much of that damage the tested methods returned. This section reads those outcomes as design evidence. The question it answers is not which existing method won, but what a mitigation method would have to read, and to do, in order to address the shortages this dataset actually presents.

5.1 What the evidence supports

On RQ1, nominal patch support and independent patient support are distinct sources of damage. Nominal deprivation reduced balanced accuracy in eight of nine comparisons, with larger losses at greater severity and when scarce support was assigned to harder classes. The affected tier followed the support allocation: when the easiest classes received the fewest patches, the body rather than the nominal tail carried the loss. Reducing patient breadth while keeping every class's patch count unchanged reduced balanced accuracy by 2.6–3.0 percentage points in all three assignments. Its probability-quality deficits of 0.132–0.135 nats were below the 0.148-nat minimum used for recovery analysis. Patch count alone therefore does not characterize whether a class has enough training evidence.

For RQ2, weighting by prevalence or nominal support recovers most of the damage caused by nominal shortage. Weighting by independent support, difficulty, or diversity recovers essentially none of the discrimination loss and worsens raw tail-group probability quality in all 24 eligible comparisons. Every deterioration remains significant after multiplicity adjustment, and 21 of the 24 remain deteriorations with intervals excluding zero after temperature scaling. The allocation-induced deficit those methods were applied to is the part that scaling mostly removes: three of the eight scaled deficits still clear the materiality minimum, and three reverse sign. Holding β constant between independent-support and class-balanced weighting does not close the discrimination gap. More importantly, none of the five confirmatory methods or nine exploratory mechanisms repairs the discrimination loss caused by reduced patient breadth. The direct advantage of matched over unmatched weighting in those conditions is only 0.06 percentage points of balanced accuracy, with an interval covering zero. These results support prevalence-based weighting for nominal shortage, but not class weighting as a remedy for limited patient evidence.

RQ3 remains pending until the TCGA-UT results and combined fit are added in Section 4.

5.2 Which shortage a mitigation method must read

A method must estimate the shortage that binds, so the empirical question is whether the dominant shortage was identifiable and whether reading it helped. On this dataset the dominant shortage was identifiable in nine of 12 units, and reading it helped only where it was nominal support. The more informative case is the one in which the dominant shortage is independent support and reading it still does not help.

That case is a real measurement rather than a structural artefact. In a design where a single per-class evidence pool serves every condition, independent support cannot vary, and a method built to read it is being tested on conditions in which the quantity it reads is constant — a null under those circumstances says nothing. The spread arm removes that objection: independent support varies by 0.41 to 0.67 nats between the two arms, the pools are nested so the manipulation adds patients rather than exchanging them, and the resulting deficit is sufficiently large and precise for recovery analysis. Independent-support weighted cross-entropy then recovers only 1.3–1.5% of it. The inference the evidence supports is that a mean-one class weight computed from patient counts does not repair damage caused by a shortage of patients, at this dose, on this dataset.

The mechanism that failure suggests is worth stating because it constrains method design directly. All five confirmatory members apply a per-class scalar to the loss, which can only change how heavily each class's examples are emphasized. When a class lacks patients rather than patches, its patches are correlated: they carry fewer independent observations of the class than their count implies, and no reweighting of that count adds information the sample does not contain. A method that addresses independent-support shortage must therefore do something other than reweight — draw on evidence outside the class's own examples, or change what is estimated rather than how heavily it is weighted.

The frequency of the ambiguous designation qualifies the routing question separately. Three of 12 units, all at severe severity, have their two largest shortage scores within a quarter of a standard deviation. A taxonomy that cannot separate the top two axes in a quarter of its conditions does not partition sharply enough to route a method by selecting among signals, which points toward a method that combines them.

5.3 Signal against mechanism

Because the five confirmatory methods share one intervention mechanism and differ only in the signal their weights read, the spread of recovery across them isolates how much leverage the choice of signal carries. That spread is wide on the nominal axis, running from below zero to 89.2% within single comparisons, so signal choice determines whether a weighted-loss method works at all there. The spread is not, however, evidence that a better shortage estimate would help, because it is bimodal rather than graded: two members work almost everywhere on that axis and three work almost nowhere, with little in between.

The independent-support axis separates signal from mechanism more cleanly than the nominal axis can, because there the mechanism is held fixed across the five members and the wider roster fails alongside them. Nine exploratory methods spanning resampling, logit adjustment, decision-threshold correction, classifier retraining, and set learning were applied to a deficit of roughly three balanced-accuracy points, and none returned a positive fraction of it whose interval excludes zero. When both the signal contrast and the mechanism contrast come back null on the same units, the limitation lies not in the choice among tested designs but in what all of them share.

The roster's harmful members sharpen that reading. Set learning and focal loss damage the independent-support comparisons most, with recovery between −64% and −106%, and classifier retraining damages them too, between −33% and −38%. All three rebalance the effective training distribution more aggressively than a per-class scalar does, whether by resampling, by re-emphasizing hard examples, or by refitting the classifier on a rebalanced head. Rebalancing toward classes whose examples come from few patients amplifies those patients' idiosyncrasies rather than the class's signal, which is the failure mode a patient-aware method would have to avoid by construction.

5.4 Where tested methods reach a ceiling

The clearest unmet needs are the four discrimination comparisons in which no confirmatory or exploratory method returns a substantial fraction of an established deficit. Three are the independent-support comparisons, where nothing recovers at all. The fourth is difficulty-aligned severe (spread), where the best tested mechanism reaches 55.4% recovery. A related failure appears in native severe (spread): both prevalence-based methods worsen tail-group negative log likelihood even though they repair most of the same condition's discrimination deficit. Post-hoc calibration does not remove it. Their raw deteriorations of −0.083 and −0.090 nats have intervals covering zero, whereas the scaled ones, −0.218 and −0.217 nats, do not.

This is the finding that most changes what a new method would be for. The ceiling on the nominal axis is set by neither the data nor the fixed representation — existing mechanisms reach most of that damage, and a new method cannot be motivated there by damage left unrepaired. The ceiling on the independent-support axis is set by the mechanism the whole roster shares. A method motivated by this benchmark is therefore a method for the second regime, and its claim would be a claim about patient-level evidence rather than about class counts.

The complementary case remains part of the specification. The difficulty-aligned moderate condition damaged raw probability quality without a certifiable discrimination deficit, and the three independent-support units damaged discrimination without a certifiable raw probability-quality deficit. A method that intervenes on the wrong axis spends capacity and risks degrading an endpoint that was not damaged, and Section 3.3 shows that risk: in all 24 cells where a non-prevalence signal was applied to a calibration-damaged unit, raw tail-group negative log likelihood rose above the untreated reference, and 21 of those 24 remain deteriorations with intervals excluding zero after temperature scaling. This harm is therefore a property of the intervention rather than a confidence-scaling artefact, which is what makes declining to intervene a requirement rather than a preference.

5.5 Design requirements and open questions

The preceding subsections resolve into a set of requirements for a mitigation method, each traceable to a specific result, annotated with the strength of evidence behind it and, where relevant, with what a second dataset would have to confirm.

1. Where the shortage is nominal, the method must read prevalence or nominal support, and must not rely on a difficulty or diversity estimate as its primary signal. *Confirmatory*, from the five-family separation in Table 4 across the eight nominal-axis discrimination units.

2. It must not leave tail-group probability quality worse after validation-fitted calibration while repairing discrimination. *Confirmatory for raw scores*, from the 24 negative recovery cells in Table 6, all of which survive Holm adjustment, and *supported after calibration* by the 21 of those 24 whose scaled intervals still exclude zero. The requirement binds on the harm rather than on the repair: the raw benefit credited to prevalence-based weighting on this endpoint does not survive scaling.

3. It must decline to intervene where allocation produced no material deficit on the axis it targets, which on this dataset is the difficulty-aligned moderate condition on discrimination and the three independent-support units on probability quality. *Confirmatory*, from Sections 2.2 and 2.3.

4. Independent-support shortage requires a mechanism other than per-class loss reweighting. *Confirmatory within this dataset*, from the joint null of the confirmatory family and the exploratory roster on the three independent-support comparisons. *Open across datasets*: this cohort carries the weakest independent dose the candidate datasets offer, so a replication at a stronger dose is needed before the null is read as a property of reweighting rather than of this dose.

5. Whether the shortage estimate must be formed per condition rather than transferred across datasets. *Pending TCGA-UT*: this is the requirement RQ3's variance components were specified to settle.

Requirements four and five will be reassessed after the TCGA-UT analysis. If the independent-support result replicates, the fastest route to a new method is a focused mechanism experiment rather than another broad method roster. Such an experiment should hold patch counts fixed while varying patient breadth and within-patient redundancy, then test a patient-aware objective instead of applying another per-class weight. A practical first candidate is a patient-macro objective, which averages patch losses within each patient before giving patients equal weight; hierarchical shrinkage that borrows strength across patients or classes is a more complex second candidate. A diversity-focused arm is justified only if the combined RQ3 analysis links residual damage to representational diversity; the present failure of diversity weighting makes it a weaker starting point. Any candidate should be judged jointly on balanced accuracy and probability quality and should remain inactive where its target deficit is absent.

5.6 Pending TCGA-UT analysis

Placeholder: this subsection will report whether the independent-support result replicates on TCGA-UT, compare dataset-specific materiality thresholds, and interpret the combined RQ3 model. Binary and low-cardinality targets remain outside that model because support–difficulty alignment over two classes is structurally fixed at ± 1.

6 Conclusion and Scope

The current BRACS results answer the first two protocol questions. Controlled allocation damages tail-class discrimination and probability quality along two axes that ordinary imbalance confounds: nominal patch deprivation and independent patient deprivation. Signal-matched mitigation recovers most nominal damage through prevalence and nominal-support weighting, but no tested signal or mechanism recovers the independent-support damage. Section 5 converts this evidence into requirements for what a mitigation method must read, what it must not degrade, and where it must decline to act. RQ3 will be completed with TCGA-UT.

Four conditions bound the current claims. The three partitions are repeated partitions of BRACS, not independent cohorts, so the intervals quantify variation across partitioning and initialization rather than across populations. The three independent-support units share one deprived allocation and differ only in their reference assignment, so they are correlated readings rather than three replications. The benchmark covers patch classification only, where each prediction unit carries its own label; at whole-slide level, reallocating labelled slides changes nominal and independent support together. Temperature scaling is reported as a diagnostic and never determines whether recovery is analysed.

A Realized Conditions and Analysis Validity

This appendix carries the execution record that Section 2 summarizes in prose. It documents the realized construction and whether the conditions required for confirmatory inference were met.

A.1 Realized construction

Table 9 reports the achieved imbalance ratio and normalized-entropy deficit for each partition and controlled condition, together with the realized patient, slide, and patch counts. Conditions falling outside the tolerance bands, $\rho \in [9, 11]$ for moderate and $\rho \in [90, 110]$ for severe, are flagged as off-target, and allocations that collapsed to balanced are flagged as degenerate and excluded from all estimands. No condition carries either flag. Patient and slide counts are unique within a realization: one patient may supply patches to more than one class, so per-class counts are not summed.

Dataset	Split	Condition	Assignment	Units	Slides	Patches	ρ	$1 - H_{\text{norm}}$	Tolerance
BRACS	0	Natural	—	103	271	217399	5.901	0.1202	—
BRACS	0	Balanced	—	85	204	16796	1.000	0.0000	—
BRACS	0	Balanced	Aligned	85	204	16796	1.000	0.0000	—
BRACS	0	Balanced (spread)	Aligned	103	209	16796	1.000	0.0000	—
BRACS	0	Moderate	Aligned	85	204	16796	10.007	0.1280	On target
BRACS	0	Severe	Aligned	85	194	16796	100.467	0.3530	On target
BRACS	0	Severe (spread)	Aligned	103	200	16796	100.467	0.3530	On target
BRACS	0	Balanced	Reversed	85	204	16796	1.000	0.0000	—
BRACS	0	Balanced (spread)	Reversed	103	209	16796	1.000	0.0000	—
BRACS	0	Moderate	Reversed	85	204	16796	10.007	0.1280	On target
BRACS	0	Severe	Reversed	85	203	16796	100.467	0.3530	On target
BRACS	0	Severe (spread)	Reversed	103	209	16796	100.467	0.3530	On target
BRACS	0	Balanced	Native	85	204	16796	1.000	0.0000	—
BRACS	0	Balanced (spread)	Native	103	209	16796	1.000	0.0000	—
BRACS	0	Moderate	Native	85	204	16796	10.007	0.1280	On target
BRACS	0	Severe	Native	85	204	16796	100.467	0.3530	On target
BRACS	0	Severe (spread)	Native	103	209	16796	100.467	0.3530	On target
BRACS	1	Natural	—	103	273	193010	5.722	0.1178	—
BRACS	1	Balanced	—	89	233	27202	1.000	0.0000	—
BRACS	1	Balanced	Aligned	89	233	27202	1.000	0.0000	—
BRACS	1	Balanced (spread)	Aligned	103	229	27202	1.000	0.0000	—
BRACS	1	Moderate	Aligned	89	233	27202	10.003	0.1280	On target
BRACS	1	Severe	Aligned	89	232	27202	100.301	0.3530	On target
BRACS	1	Severe (spread)	Aligned	103	229	27202	100.301	0.3530	On target
BRACS	1	Balanced	Reversed	89	233	27202	1.000	0.0000	—
BRACS	1	Balanced (spread)	Reversed	103	229	27202	1.000	0.0000	—
BRACS	1	Moderate	Reversed	89	233	27202	10.003	0.1280	On target
BRACS	1	Severe	Reversed	89	233	27202	100.301	0.3530	On target
BRACS	1	Severe (spread)	Reversed	103	228	27202	100.301	0.3530	On target
BRACS	1	Balanced	Native	89	233	27202	1.000	0.0000	—
BRACS	1	Balanced (spread)	Native	103	229	27202	1.000	0.0000	—
BRACS	1	Moderate	Native	89	233	27202	10.003	0.1280	On target
BRACS	1	Severe	Native	89	233	27202	100.301	0.3530	On target
BRACS	1	Severe (spread)	Native	103	229	27202	100.301	0.3530	On target
BRACS	2	Natural	—	103	262	217704	8.297	0.1432	—
BRACS	2	Balanced	—	78	179	13982	1.001	0.0000	—
BRACS	2	Balanced	Aligned	78	179	13982	1.001	0.0000	—
BRACS	2	Balanced (spread)	Aligned	103	193	13982	1.001	0.0000	—
BRACS	2	Moderate	Aligned	78	179	13982	10.004	0.1280	On target
BRACS	2	Severe	Aligned	78	173	13982	100.360	0.3530	On target
BRACS	2	Severe (spread)	Aligned	103	187	13982	100.360	0.3530	On target
BRACS	2	Balanced	Reversed	78	179	13982	1.001	0.0000	—
BRACS	2	Balanced (spread)	Reversed	103	193	13982	1.001	0.0000	—

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Dataset	Split	Condition	Assignment	Units	Slides	Patches	ρ	$1 - H_{\mathrm{norm}}$	Tolerance
BRACS	2	Moderate	Reversed	78	179	13982	10.004	0.1280	On target
BRACS	2	Severe	Reversed	78	179	13982	100.360	0.3530	On target
BRACS	2	Severe (spread)	Reversed	103	193	13982	100.360	0.3530	On target
BRACS	2	Balanced	Native	78	179	13982	1.001	0.0000	—
BRACS	2	Balanced (spread)	Native	103	193	13982	1.001	0.0000	—
BRACS	2	Moderate	Native	78	179	13982	10.004	0.1280	On target
BRACS	2	Severe	Native	78	179	13982	100.360	0.3530	On target
BRACS	2	Severe (spread)	Native	103	193	13982	100.360	0.3530	On target

Table 9: Realized training support per partition and controlled condition: unique patients, slides, patches, achieved ρ, and normalized-entropy deficit, with tolerance status. The natural anchor and the two balanced references carry no severity tolerance.

The two rows that define the independent-support axis are the balanced and balanced (spread) entries within each partition. They carry identical patch counts and an identical ρ of 1.000, and differ only in patient count, 85 against 103, and correspondingly in slides. That the spread arm reaches the same 103 patients the natural condition draws on confirms that it exhausts the eligible inventory, so the dose reported in Section 2.1 is the largest this cohort can supply under the contribution caps.

A.2 Resampling validity and completeness

The label-only preflight resamples patients before any model is fitted and checks whether class-level endpoints can be estimated stably at all. Three concentration checks apply per class: whether the 2.5th percentile of the Kish effective count falls below five, whether a single patient carries more than half the weight in more than 5% of replicates, and whether a pooled cell draws on fewer than five contributing patients. A class failing any check designates the dataset descriptive-only, which forces both gates closed.

Dataset	Class	Kish n_{eff}	2.5th pct.	Max unit weight	Dominant-unit share	Descriptive-only
BRACS	ADH	10.8	7.2	0.180	0.000	No
BRACS	DCIS	8.8	5.5	0.211	0.000	No
BRACS	FEA	9.3	6.0	0.201	0.000	No
BRACS	IC	14.3	9.7	0.145	0.000	No
BRACS	N	14.4	9.9	0.144	0.000	No
BRACS	PB	18.9	13.5	0.116	0.000	No
BRACS	UDH	17.9	12.6	0.121	0.000	No

Table 11: Label-only bootstrap preflight per class: Kish effective count and its 2.5th percentile, maximum single-patient weight fraction, share of replicates with a dominant patient, and the resulting descriptive-only designation. Values are the most adverse across the three partitions.

No class is designated descriptive-only on the pooled evidence cell that every gated quantity uses. The tightest margin is ductal carcinoma in situ, whose 2.5th-percentile Kish effective count is 5.5 against a floor of five, and no class records a dominant patient in any replicate. The margin is narrow because the cohort is small: the seven classes draw on 85 to 103 patients in total, which is the same scarcity the independent-support axis manipulates.

That scarcity binds one level down. Evaluated per partition and per class rather than pooled, the preflight flags the ADH, DCIS, and FEA cells as descriptive-only on all three partitions, with 2.5th-percentile Kish effective counts between 1.8 and 2.7 and dominant-patient shares up to 0.32. No gated quantity in this report is computed on those cells, since every gate and every recovery estimate reads the pooled cell, but the per-class, per-partition entries of Appendix D rest on them and are descriptive for that reason.

Completeness is enforced rather than reported after the fact. Every confirmation block must contain all five initialization seeds, every expected combination of tail assignment, condition, method, and gate must appear in all three partitions, and the achieved severity of a condition must not vary by more than half across partitions.

Dataset	Splits	Seeds per block	Comparisons	Max severity spread	Descriptive-only	Guards
BRACS	3	5	378	0.002	No	All passed

Table 12: Completeness and consistency guards: confirmation-seed blocks, expected comparison coverage across the three partitions, and achieved-severity spread. These guards halt aggregation rather than degrade it, so a written result implies every guard passed.

A.3 Budget, tuning selections, and deviations

Training budget is expressed in example presentations rather than optimizer updates, because methods consume different amounts of data per update and an update budget would confound the method comparison with data exposure. The frozen budget is $E = 503{,}880$ example presentations for every controlled condition and $E = 6{,}521{,}970$ for the natural anchor; each method derives its own update count as $\lceil E/e_m \rceil$, where e_m is its examples per update. The checkpoint interval is rescaled to that derived count so that every method receives approximately 170 validation passes regardless of how many updates its budget yields.

Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Balanced	—	Balanced sampling	0.0001	0.7500	No	503880
BRACS	Balanced	—	CE	3e-05	—	No	503880
BRACS	Balanced	—	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Balanced	—	CE + soft MCC	0.0003	64.0000	No	503880
BRACS	Balanced	—	CFAL	0.0001	0.2500	No	503880
BRACS	Balanced	—	Nominal support	3e-05	0.9999	No	503880
BRACS	Balanced	—	cRT	0.003	—	No	503880
BRACS	Balanced	—	Focal loss	0.0003	1.0000	No	503880
BRACS	Balanced	—	Independent support	3e-05	0.8000	No	503880
BRACS	Balanced	—	Logit adjustment (train)	3e-05	2.0000	No	503880
BRACS	Balanced	—	OKO	0.003	2.0000	No	503880
BRACS	Balanced	—	Difficulty	3e-05	0.2500	No	503880
BRACS	Balanced	—	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Balanced	—	Diversity	3e-05	1.0000	No	503880
BRACS	Balanced	—	Prevalence	3e-05	0.7500	No	503880
BRACS	Balanced (spread)	Aligned	Balanced sampling	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	CE	3e-05	—	No	503880
BRACS	Balanced (spread)	Aligned	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Aligned	CE + soft MCC	0.0001	16.0000	No	503880
BRACS	Balanced (spread)	Aligned	CFAL	0.001	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	Nominal support	3e-05	0.9990	No	503880
BRACS	Balanced (spread)	Aligned	cRT	0.003	—	No	503880
BRACS	Balanced (spread)	Aligned	Focal loss	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Aligned	Independent support	3e-05	0.8000	No	503880
BRACS	Balanced (spread)	Aligned	Logit adjustment (train)	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	OKO	0.003	1.0000	No	503880
BRACS	Balanced (spread)	Aligned	Difficulty	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Balanced (spread)	Aligned	Diversity	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Aligned	Prevalence	3e-05	0.7500	No	503880
BRACS	Balanced (spread)	Reversed	Balanced sampling	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	CE	3e-05	—	No	503880
BRACS	Balanced (spread)	Reversed	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Reversed	CE + soft MCC	0.0001	16.0000	No	503880
BRACS	Balanced (spread)	Reversed	CFAL	0.001	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	Nominal support	3e-05	0.9990	No	503880
BRACS	Balanced (spread)	Reversed	cRT	0.003	—	No	503880
BRACS	Balanced (spread)	Reversed	Focal loss	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Reversed	Independent support	3e-05	0.8000	No	503880
BRACS	Balanced (spread)	Reversed	Logit adjustment (train)	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	OKO	0.003	1.0000	No	503880
BRACS	Balanced (spread)	Reversed	Difficulty	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	Logit adjustment (post-hoc)	—	1.0000	No	503880

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Balanced (spread)	Reversed	Diversity	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Reversed	Prevalence	3e-05	0.7500	No	503880
BRACS	Balanced (spread)	Native	Balanced sampling	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Native	CE	3e-05	—	No	503880
BRACS	Balanced (spread)	Native	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Native	CE + soft MCC	0.0001	16.0000	No	503880
BRACS	Balanced (spread)	Native	CFAL	0.001	0.2500	No	503880
BRACS	Balanced (spread)	Native	Nominal support	3e-05	0.9990	No	503880
BRACS	Balanced (spread)	Native	cRT	0.003	—	No	503880
BRACS	Balanced (spread)	Native	Focal loss	3e-05	0.0000	No	503880
BRACS	Balanced (spread)	Native	Independent support	3e-05	0.8000	No	503880
BRACS	Balanced (spread)	Native	Logit adjustment (train)	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Native	OKO	0.003	1.0000	No	503880
BRACS	Balanced (spread)	Native	Difficulty	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Native	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Balanced (spread)	Native	Diversity	3e-05	0.2500	No	503880
BRACS	Balanced (spread)	Native	Prevalence	3e-05	0.7500	No	503880
BRACS	Moderate	Aligned	Balanced sampling	0.0001	1.0000	No	503880
BRACS	Moderate	Aligned	CE	0.003	—	No	503880
BRACS	Moderate	Aligned	CE + soft F_1	0.0001	0.0000	No	503880
BRACS	Moderate	Aligned	CE + soft MCC	0.0001	0.0000	No	503880
BRACS	Moderate	Aligned	CFAL	0.003	0.2500	No	503880
BRACS	Moderate	Aligned	Nominal support	0.0001	0.9999	No	503880
BRACS	Moderate	Aligned	cRT	0.01	—	No	503880
BRACS	Moderate	Aligned	Focal loss	0.0001	0.5000	No	503880
BRACS	Moderate	Aligned	Independent support	0.001	0.9900	No	503880
BRACS	Moderate	Aligned	Logit adjustment (train)	0.0001	1.0000	No	503880
BRACS	Moderate	Aligned	OKO	0.001	4.0000	No	503880
BRACS	Moderate	Aligned	Difficulty	0.001	0.2500	No	503880
BRACS	Moderate	Aligned	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Moderate	Aligned	Diversity	0.003	1.0000	No	503880
BRACS	Moderate	Aligned	Prevalence	0.0001	1.0000	No	503880
BRACS	Moderate	Reversed	Balanced sampling	0.0001	1.0000	No	503880
BRACS	Moderate	Reversed	CE	0.003	—	No	503880
BRACS	Moderate	Reversed	CE + soft F_1	0.0001	0.0000	No	503880
BRACS	Moderate	Reversed	CE + soft MCC	0.0001	0.0000	No	503880
BRACS	Moderate	Reversed	CFAL	0.003	0.2500	No	503880
BRACS	Moderate	Reversed	Nominal support	0.0001	0.9999	No	503880
BRACS	Moderate	Reversed	cRT	0.01	—	No	503880
BRACS	Moderate	Reversed	Focal loss	0.0001	0.5000	No	503880
BRACS	Moderate	Reversed	Independent support	0.001	0.9900	No	503880
BRACS	Moderate	Reversed	Logit adjustment (train)	0.0001	1.0000	No	503880
BRACS	Moderate	Reversed	OKO	0.001	4.0000	No	503880

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Moderate	Reversed	Difficulty	0.001	0.2500	No	503880
BRACS	Moderate	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Moderate	Reversed	Diversity	0.003	1.0000	No	503880
BRACS	Moderate	Reversed	Prevalence	0.0001	1.0000	No	503880
BRACS	Moderate	Native	Balanced sampling	0.0001	1.0000	No	503880
BRACS	Moderate	Native	CE	0.003	—	No	503880
BRACS	Moderate	Native	CE + soft F_1	0.0001	0.0000	No	503880
BRACS	Moderate	Native	CE + soft MCC	0.0001	0.0000	No	503880
BRACS	Moderate	Native	CFAL	0.003	0.2500	No	503880
BRACS	Moderate	Native	Nominal support	0.0001	0.9999	No	503880
BRACS	Moderate	Native	cRT	0.01	—	No	503880
BRACS	Moderate	Native	Focal loss	0.0001	0.5000	No	503880
BRACS	Moderate	Native	Independent support	0.001	0.9900	No	503880
BRACS	Moderate	Native	Logit adjustment (train)	0.0001	1.0000	No	503880
BRACS	Moderate	Native	OKO	0.001	4.0000	No	503880
BRACS	Moderate	Native	Difficulty	0.001	0.2500	No	503880
BRACS	Moderate	Native	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Moderate	Native	Diversity	0.003	1.0000	No	503880
BRACS	Moderate	Native	Prevalence	0.0001	1.0000	No	503880
BRACS	Natural	—	CE	3e-05	—	No	6521970
BRACS	Severe	Aligned	Balanced sampling	1e-05	1.0000	No	503880
BRACS	Severe	Aligned	CE	0.001	—	No	503880
BRACS	Severe	Aligned	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe	Aligned	CE + soft MCC	1e-05	0.0000	No	503880
BRACS	Severe	Aligned	CFAL	0.003	1.0000	No	503880
BRACS	Severe	Aligned	Nominal support	0.0003	0.9999	No	503880
BRACS	Severe	Aligned	cRT	0.01	—	No	503880
BRACS	Severe	Aligned	Focal loss	3e-05	0.0000	No	503880
BRACS	Severe	Aligned	Independent support	0.001	0.9990	No	503880
BRACS	Severe	Aligned	Logit adjustment (train)	0.0003	1.0000	No	503880
BRACS	Severe	Aligned	OKO	0.001	4.0000	No	503880
BRACS	Severe	Aligned	Difficulty	0.001	0.5000	No	503880
BRACS	Severe	Aligned	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Severe	Aligned	Diversity	0.003	0.5000	No	503880
BRACS	Severe	Aligned	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe	Reversed	Balanced sampling	1e-05	1.0000	No	503880
BRACS	Severe	Reversed	CE	0.001	—	No	503880
BRACS	Severe	Reversed	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe	Reversed	CE + soft MCC	1e-05	0.0000	No	503880
BRACS	Severe	Reversed	CFAL	0.003	1.0000	No	503880
BRACS	Severe	Reversed	Nominal support	0.0003	0.9999	No	503880
BRACS	Severe	Reversed	cRT	0.01	—	No	503880
BRACS	Severe	Reversed	Focal loss	3e-05	0.0000	No	503880

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Severe	Reversed	Independent support	0.001	0.9990	No	503880
BRACS	Severe	Reversed	Logit adjustment (train)	0.0003	1.0000	No	503880
BRACS	Severe	Reversed	OKO	0.001	4.0000	No	503880
BRACS	Severe	Reversed	Difficulty	0.001	0.5000	No	503880
BRACS	Severe	Reversed	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Severe	Reversed	Diversity	0.003	0.5000	No	503880
BRACS	Severe	Reversed	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe	Native	Balanced sampling	1e-05	1.0000	No	503880
BRACS	Severe	Native	CE	0.001	—	No	503880
BRACS	Severe	Native	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe	Native	CE + soft MCC	1e-05	0.0000	No	503880
BRACS	Severe	Native	CFAL	0.003	1.0000	No	503880
BRACS	Severe	Native	Nominal support	0.0003	0.9999	No	503880
BRACS	Severe	Native	cRT	0.01	—	No	503880
BRACS	Severe	Native	Focal loss	3e-05	0.0000	No	503880
BRACS	Severe	Native	Independent support	0.001	0.9990	No	503880
BRACS	Severe	Native	Logit adjustment (train)	0.0003	1.0000	No	503880
BRACS	Severe	Native	OKO	0.001	4.0000	No	503880
BRACS	Severe	Native	Difficulty	0.001	0.5000	No	503880
BRACS	Severe	Native	Logit adjustment (post-hoc)	—	0.5000	No	503880
BRACS	Severe	Native	Diversity	0.003	0.5000	No	503880
BRACS	Severe	Native	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Balanced sampling	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Aligned	CE	0.001	—	No	503880
BRACS	Severe (spread)	Aligned	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Aligned	CE + soft MCC	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Aligned	CFAL	0.001	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Nominal support	0.0001	0.9999	No	503880
BRACS	Severe (spread)	Aligned	cRT	0.01	—	No	503880
BRACS	Severe (spread)	Aligned	Focal loss	3e-05	0.5000	No	503880
BRACS	Severe (spread)	Aligned	Independent support	0.001	0.9900	No	503880
BRACS	Severe (spread)	Aligned	Logit adjustment (train)	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Aligned	OKO	0.0003	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Difficulty	0.001	0.2500	No	503880
BRACS	Severe (spread)	Aligned	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Diversity	0.001	1.0000	No	503880
BRACS	Severe (spread)	Aligned	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Balanced sampling	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Reversed	CE	0.001	—	No	503880
BRACS	Severe (spread)	Reversed	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Reversed	CE + soft MCC	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Reversed	CFAL	0.001	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Nominal support	0.0001	0.9999	No	503880

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Dataset	Condition	Assignment	Method	Learning rate	Control	At envelope bound	Example presentations
BRACS	Severe (spread)	Reversed	cRT	0.01	—	No	503880
BRACS	Severe (spread)	Reversed	Focal loss	3e-05	0.5000	No	503880
BRACS	Severe (spread)	Reversed	Independent support	0.001	0.9900	No	503880
BRACS	Severe (spread)	Reversed	Logit adjustment (train)	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Reversed	OKO	0.0003	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Difficulty	0.001	0.2500	No	503880
BRACS	Severe (spread)	Reversed	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Diversity	0.001	1.0000	No	503880
BRACS	Severe (spread)	Reversed	Prevalence	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Native	Balanced sampling	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Native	CE	0.001	—	No	503880
BRACS	Severe (spread)	Native	CE + soft F_1	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Native	CE + soft MCC	3e-05	0.0000	No	503880
BRACS	Severe (spread)	Native	CFAL	0.001	1.0000	No	503880
BRACS	Severe (spread)	Native	Nominal support	0.0001	0.9999	No	503880
BRACS	Severe (spread)	Native	cRT	0.01	—	No	503880
BRACS	Severe (spread)	Native	Focal loss	3e-05	0.5000	No	503880
BRACS	Severe (spread)	Native	Independent support	0.001	0.9900	No	503880
BRACS	Severe (spread)	Native	Logit adjustment (train)	3e-05	1.0000	No	503880
BRACS	Severe (spread)	Native	OKO	0.0003	1.0000	No	503880
BRACS	Severe (spread)	Native	Difficulty	0.001	0.2500	No	503880
BRACS	Severe (spread)	Native	Logit adjustment (post-hoc)	—	1.0000	No	503880
BRACS	Severe (spread)	Native	Diversity	0.001	1.0000	No	503880
BRACS	Severe (spread)	Native	Prevalence	3e-05	1.0000	No	503880

Table 14: Selected learning rate and strength control per condition, tail assignment, and method, with the frozen example-presentation budget and whether the selection terminated at a learning-rate envelope boundary.

None of the 196 selections terminated at a learning-rate envelope boundary, so in no case did the envelope rather than the validation data determine the choice. Three properties of the selection procedure are stated here as limits on the claims that rest on it rather than as defects.

The selection objective is natural-prevalence validation balanced accuracy, tie-broken by macro F_1 and then by negative log likelihood. Tail-group negative log likelihood is a co-primary endpoint with its own gate, and no method was ever tuned for it. Every statement in Section 3.3 about a method degrading probability quality is therefore a statement about a method optimized for a different endpoint.

Selection also pools across tail assignments: one configuration is chosen per condition and method and applied to all three assignments, which the table records as repeated rows, and all 60 condition–method cells that span more than one assignment carry identical selections. Since the assignment determines which classes need help, this pooling systematically dilutes the difficulty- and diversity-reading methods relative to a per-assignment selection, and their null results should be read with that in mind.

The tuning budget itself differs by method: 16 candidates for most, four for cross-entropy and classifier retraining, and a single shard for post-hoc logit adjustment, which requires no training of its own.

Three further properties of the selections change what a method label denotes and are recorded for that reason. The search selects the zero-strength point in 28 of the 196 selections: 13 for the soft-F_1 hybrid, nine for the soft-MCC hybrid, and six for focal loss. At zero strength the soft hybrids reduce to balanced-sampling cross-entropy, because their auxiliary term vanishes while their forced balanced sampler does not, and focal loss reduces to class-weighted cross-entropy, because its class-aware α term applies at unit strength independently of γ. Where those selections were made, the roster row reports the reduced method rather than the named one.

That reduction previously interacted with a defect in the search. The search treated focal loss as anchored to plain cross-entropy, scoring its $\gamma = 0$ candidate with cross-entropy's already-computed validation metrics instead of training it; because α applies unconditionally, that candidate is class-weighted cross-entropy rather than plain cross-entropy, so the anchor supplied the wrong metrics. The anchor has been removed, and the two conditions whose focal selection had come from it, concentrated balanced and balanced spread, were re-tuned from the first round under the corrected search. Concentrated balanced now selects $\gamma = 1$; balanced spread again selects $\gamma = 0$, this time against a trained candidate, as does concentrated severe, whose window had already reached $\gamma = 0$ under the original search. Every zero-strength focal selection reported here therefore rests on its own fitted evidence. Re-tuning those two conditions also moved four other selections in them, which is why their estimates differ from a search that never trained a $\gamma = 0$ point.

Two deviations from the locked protocol were made before any definitive gate row was read. The two deficit-gate thresholds were derived rather than assumed: the protocol had fixed discrimination at 2 percentage points and calibration at 0.05 nats without a stated derivation, while the discrimination value conflicted with the support pilot's 1-percentage-point materiality constant for the same endpoint under the same kind of manipulation. Both were replaced by quantities the protocol already fixes: $\max(1, 2\sigma_{\text{seed}}) = 1.443$ percentage points for discrimination and $2\sigma_{\text{seed}} = 0.14782$ nats for calibration, where σ_{seed} is the confirmation-seed standard deviation of the corresponding balanced-condition endpoint on this dataset. Separately, the gate-defining estimator is case-macro rather than patch-micro, and the evidence cell is pooled across a case's partition appearances rather than taken per partition; both correct a mismatch between a diagnostic and the estimator it certifies, since the bootstrap draws one shared weight per case over the union of the three test frames.

A.4 Method diagnostics

Semantic-scale weighted cross-entropy falls back to a raw weight of 1, that is, to no reweighting, for any class whose matched draw is degenerate at reweighting time. The `ssb_invalid_draws` counter records how often that fallback fires; a nonzero count would be grounds to distrust the diversity-weighted recovery readings for the affected condition.

Dataset	Split	Condition	Method	Seeds	Ssb invalid draws	Ssb pools
BRACS	0	Balanced (spread)	Diversity	15	0	15
BRACS	0	Moderate	Diversity	15	0	15
BRACS	0	Severe	Diversity	15	0	15
BRACS	0	Severe (spread)	Diversity	15	0	15
BRACS	0	Balanced	Diversity	5	0	5
BRACS	1	Balanced (spread)	Diversity	15	0	15
BRACS	1	Moderate	Diversity	15	0	15
BRACS	1	Severe	Diversity	15	0	15
BRACS	1	Severe (spread)	Diversity	15	0	15
BRACS	1	Balanced	Diversity	5	0	5
BRACS	2	Balanced (spread)	Diversity	15	0	15
BRACS	2	Moderate	Diversity	15	0	15
BRACS	2	Severe	Diversity	15	0	15
BRACS	2	Severe (spread)	Diversity	15	0	15
BRACS	2	Balanced	Diversity	5	0	5

Table 16: Per-partition, per-condition `method_diagnostics` counters for semantic-scale weighted cross-entropy: degenerate-draw fallback count against the number of reweighting pools formed, summed across confirmation seeds.

The fallback never fired: 0 invalid draws against 195 reweighting pools across the three partitions. The diversity-weighted results in Section 3 therefore measure the intended weighting rather than a degenerate fallback.

B Aggregate Damage Estimates

Tables 17 and 18 provide the assignment-level estimates summarized in Sections 2.2 and 2.3.

Dataset	Assignment	Severity	D_{BA} (pp)	95% CI (pp)	Gate
BRACS	Native	Balanced	2.91	[1.02, 5.19]	Open
BRACS	Aligned	Balanced	3.03	[1.02, 5.31]	Open
BRACS	Reversed	Balanced	2.62	[0.53, 5.02]	Open
BRACS	Native	Moderate	3.34	[1.72, 5.09]	Open
BRACS	Aligned	Moderate	1.50	[-0.40, 3.39]	Interval covers zero
BRACS	Reversed	Moderate	8.17	[6.09, 10.60]	Open
BRACS	Native	Severe	5.45	[3.27, 7.81]	Open
BRACS	Aligned	Severe	4.30	[2.08, 6.59]	Open
BRACS	Reversed	Severe	13.23	[10.78, 15.84]	Open
BRACS	Native	Severe (spread)	8.43	[5.79, 11.37]	Open
BRACS	Aligned	Severe (spread)	4.15	[1.97, 6.62]	Open
BRACS	Reversed	Severe (spread)	13.77	[11.33, 16.34]	Open

Table 17: Balanced-accuracy deficit with paired 95% intervals.

Dataset	Assignment	Severity	D_{cal} (nats)	95% CI	Gate	ECE	ECE 95% CI
BRACS	Native	Balanced	0.138	[0.064, 0.216]	Below threshold	0.051	[0.040, 0.087]
BRACS	Aligned	Balanced	0.140	[0.071, 0.213]	Below threshold	0.051	[0.040, 0.087]
BRACS	Reversed	Balanced	0.134	[0.061, 0.209]	Below threshold	0.051	[0.040, 0.087]
BRACS	Native	Moderate	0.186	[-0.007, 0.397]	Interval covers zero	0.148	[0.103, 0.210]
BRACS	Aligned	Moderate	0.379	[0.157, 0.715]	Open	0.143	[0.108, 0.189]
BRACS	Reversed	Moderate	0.754	[0.180, 1.706]	Open	0.215	[0.170, 0.276]
BRACS	Native	Severe	0.688	[0.332, 1.099]	Open	0.220	[0.176, 0.276]
BRACS	Aligned	Severe	0.729	[0.488, 1.045]	Open	0.219	[0.174, 0.280]
BRACS	Reversed	Severe	1.744	[1.208, 2.295]	Open	0.313	[0.256, 0.384]
BRACS	Native	Severe (spread)	0.205	[0.100, 0.334]	Open	0.147	[0.112, 0.198]
BRACS	Aligned	Severe (spread)	0.908	[0.575, 1.267]	Open	0.209	[0.160, 0.275]
BRACS	Reversed	Severe (spread)	1.129	[0.830, 1.474]	Open	0.259	[0.208, 0.327]

Table 18: Tail-group negative-log-likelihood deficit with paired 95% intervals and secondary expected calibration error.

C Detailed Recovery Estimates

The main text retains only the comparisons needed to answer RQ2. This appendix provides every condition-level estimate, paired bootstrap interval, and multiplicity-adjusted test result. Recovery is shown as a fraction here, so 1 corresponds to 100% in the main-text summaries. Bootstrap intervals quantify patient-resampling uncertainty, whereas the partition-block permutation tests determine the confirmatory decisions.

Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status
BRACS	Native	Balanced	Prevalence	0.003	[-0.014, 0.031]	0.5728	1.0000	Tested
BRACS	Native	Balanced	Nominal support	0.005	[-0.013, 0.033]	0.3891	1.0000	Tested
BRACS	Native	Balanced	Independent support	0.013	[-0.008, 0.055]	0.1222	1.0000	Tested
BRACS	Native	Balanced	Difficulty	-0.058	[-0.256, 0.066]	0.3557	1.0000	Tested
BRACS	Native	Balanced	Diversity	0.022	[-0.230, 0.300]	0.7409	1.0000	Tested
BRACS	Aligned	Balanced	Prevalence	0.003	[-0.014, 0.030]	0.5728	1.0000	Tested
BRACS	Aligned	Balanced	Nominal support	0.004	[-0.012, 0.032]	0.3891	1.0000	Tested
BRACS	Aligned	Balanced	Independent support	0.013	[-0.008, 0.052]	0.1222	1.0000	Tested
BRACS	Aligned	Balanced	Difficulty	-0.056	[-0.249, 0.062]	0.3557	1.0000	Tested
BRACS	Aligned	Balanced	Diversity	0.021	[-0.223, 0.281]	0.7409	1.0000	Tested
BRACS	Reversed	Balanced	Prevalence	0.003	[-0.021, 0.040]	0.5728	1.0000	Tested
BRACS	Reversed	Balanced	Nominal support	0.005	[-0.018, 0.043]	0.3891	1.0000	Tested
BRACS	Reversed	Balanced	Independent support	0.015	[-0.010, 0.082]	0.1222	1.0000	Tested
BRACS	Reversed	Balanced	Difficulty	-0.065	[-0.372, 0.081]	0.3557	1.0000	Tested
BRACS	Reversed	Balanced	Diversity	0.025	[-0.341, 0.347]	0.7409	1.0000	Tested
BRACS	Native	Moderate	Prevalence	0.815	[0.419, 1.237]	0.0002	0.0130	Tested
BRACS	Native	Moderate	Nominal support	0.892	[0.549, 1.307]	<0.0001	0.0022	Tested
BRACS	Native	Moderate	Independent support	0.197	[-0.190, 0.557]	0.1928	1.0000	Tested
BRACS	Native	Moderate	Difficulty	0.121	[-0.263, 0.453]	0.4373	1.0000	Tested
BRACS	Native	Moderate	Diversity	0.142	[-0.415, 0.622]	0.2702	1.0000	Tested
BRACS	Reversed	Moderate	Prevalence	0.840	[0.720, 0.962]	<0.0001	0.0026	Tested
BRACS	Reversed	Moderate	Nominal support	0.833	[0.696, 0.969]	<0.0001	0.0011	Tested
BRACS	Reversed	Moderate	Independent support	0.051	[-0.144, 0.202]	0.5917	1.0000	Tested
BRACS	Reversed	Moderate	Difficulty	-0.078	[-0.244, 0.059]	0.2205	1.0000	Tested
BRACS	Reversed	Moderate	Diversity	-0.064	[-0.279, 0.126]	0.2047	1.0000	Tested
BRACS	Native	Severe	Prevalence	0.768	[0.441, 1.030]	0.0025	0.1214	Tested
BRACS	Native	Severe	Nominal support	0.637	[0.353, 0.880]	0.0028	0.1297	Tested
BRACS	Native	Severe	Independent support	0.106	[-0.149, 0.323]	0.2974	1.0000	Tested
BRACS	Native	Severe	Difficulty	-0.273	[-0.637, -0.070]	0.0012	0.0603	Tested
BRACS	Native	Severe	Diversity	-0.165	[-0.548, 0.049]	0.0280	1.0000	Tested
BRACS	Aligned	Severe	Prevalence	0.654	[0.330, 1.061]	0.0073	0.3344	Tested
BRACS	Aligned	Severe	Nominal support	0.426	[0.002, 0.767]	0.0614	1.0000	Tested
BRACS	Aligned	Severe	Independent support	0.024	[-0.193, 0.214]	0.7576	1.0000	Tested
BRACS	Aligned	Severe	Difficulty	0.164	[-0.071, 0.437]	0.0361	1.0000	Tested
BRACS	Aligned	Severe	Diversity	-0.059	[-0.524, 0.237]	0.5979	1.0000	Tested
BRACS	Reversed	Severe	Prevalence	0.794	[0.691, 0.889]	<0.0001	0.0006	Tested
BRACS	Reversed	Severe	Nominal support	0.736	[0.619, 0.856]	<0.0001	0.0006	Tested

Continued on next page

Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status
BRACS	Reversed	Severe	Independent support	0.111	[0.046, 0.184]	0.0010	0.0505	Tested
BRACS	Reversed	Severe	Difficulty	-0.059	[-0.120, 0.002]	0.0185	0.8122	Tested
BRACS	Reversed	Severe	Diversity	-0.010	[-0.092, 0.064]	0.7536	1.0000	Tested
BRACS	Native	Severe (spread)	Prevalence	0.871	[0.709, 1.058]	$<$0.0001	0.0016	Tested
BRACS	Native	Severe (spread)	Nominal support	0.831	[0.675, 1.013]	$<$0.0001	0.0006	Tested
BRACS	Native	Severe (spread)	Independent support	0.111	[-0.033, 0.227]	0.0411	1.0000	Tested
BRACS	Native	Severe (spread)	Difficulty	0.021	[-0.173, 0.190]	0.9236	1.0000	Tested
BRACS	Native	Severe (spread)	Diversity	-0.026	[-0.151, 0.074]	0.5866	1.0000	Tested
BRACS	Aligned	Severe (spread)	Prevalence	0.466	[0.005, 0.823]	0.0517	1.0000	Tested
BRACS	Aligned	Severe (spread)	Nominal support	0.455	[0.085, 0.755]	0.0297	1.0000	Tested
BRACS	Aligned	Severe (spread)	Independent support	-0.135	[-0.461, 0.059]	0.0738	1.0000	Tested
BRACS	Aligned	Severe (spread)	Difficulty	-0.145	[-0.836, 0.204]	0.1362	1.0000	Tested
BRACS	Aligned	Severe (spread)	Diversity	-0.145	[-0.616, 0.112]	0.0831	1.0000	Tested
BRACS	Reversed	Severe (spread)	Prevalence	0.824	[0.714, 0.940]	$<$0.0001	0.0006	Tested
BRACS	Reversed	Severe (spread)	Nominal support	0.791	[0.685, 0.906]	$<$0.0001	0.0006	Tested
BRACS	Reversed	Severe (spread)	Independent support	0.051	[-0.006, 0.113]	0.0204	0.8772	Tested
BRACS	Reversed	Severe (spread)	Difficulty	-0.008	[-0.091, 0.059]	0.6915	1.0000	Tested
BRACS	Reversed	Severe (spread)	Diversity	0.030	[-0.037, 0.092]	0.1580	1.0000	Tested

Table 20: Confirmatory balanced-accuracy recovery by condition and weighting signal.

Dataset	Assignment	Severity	Gate	Dominant	Contrast	95% CI (bootstrap)	Holm p	Best recoverer	Agrees
BRACS	Native	Balanced	Discrimination	Independent support	0.0006	[-0.0013, 0.0022]	1.0000	Diversity	No
BRACS	Aligned	Balanced	Discrimination	Independent support	0.0006	[-0.0013, 0.0022]	1.0000	Diversity	No
BRACS	Reversed	Balanced	Discrimination	Independent support	0.0006	[-0.0013, 0.0022]	1.0000	Diversity	No
BRACS	Native	Moderate	Discrimination	Nominal support	0.0234	[0.0096, 0.0375]	0.0367	Nominal support	Yes
BRACS	Reversed	Moderate	Discrimination	Nominal support	0.0708	[0.0525, 0.0915]	0.0006	Prevalence	Yes
BRACS	Native	Severe	Discrimination	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Severe	Discrimination	Difficulty	-0.0042	[-0.0163, 0.0086]	1.0000	Prevalence	No
BRACS	Reversed	Severe	Discrimination	Ambiguous	—	—	—	Prevalence	—
BRACS	Native	Severe (spread)	Discrimination	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Severe (spread)	Discrimination	Difficulty	-0.0127	[-0.0242, 0.0005]	0.4554	Prevalence	No
BRACS	Reversed	Severe (spread)	Discrimination	Diversity	-0.0529	[-0.0648, -0.0418]	0.0006	Prevalence	No
BRACS	Aligned	Moderate	Calibration	Difficulty	-0.5566	[-1.4624, 0.0712]	0.0004	Nominal support	No
BRACS	Reversed	Moderate	Calibration	Nominal support	1.3341	[1.0771, 1.5953]	0.0004	Prevalence	Yes
BRACS	Native	Severe	Calibration	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Severe	Calibration	Difficulty	-0.4862	[-0.8830, -0.1232]	0.0004	Prevalence	No
BRACS	Reversed	Severe	Calibration	Ambiguous	—	—	—	Prevalence	—
BRACS	Native	Severe (spread)	Calibration	Ambiguous	—	—	—	Prevalence	—
BRACS	Aligned	Severe (spread)	Calibration	Difficulty	-0.3229	[-0.5781, -0.1043]	0.0004	Nominal support	No
BRACS	Reversed	Severe (spread)	Calibration	Diversity	-0.7637	[-1.0321, -0.5357]	0.0004	Prevalence	No

Table 22: Matched-minus-unmatched contrasts for balanced accuracy and probability quality.

Dataset	Assignment	Severity	CE	Independent	Independent at matched β	Nominal
BRACS	Native	Balanced	0.492	0.493	0.484	0.492
BRACS	Aligned	Balanced	0.492	0.493	0.484	0.492
BRACS	Reversed	Balanced	0.492	0.493	0.484	0.492
BRACS	Native	Moderate	0.467	0.479	0.476	0.495
BRACS	Aligned	Moderate	0.460	0.456	0.457	0.477
BRACS	Reversed	Moderate	0.423	0.434	0.431	0.486
BRACS	Native	Severe	0.447	0.456	0.454	0.479
BRACS	Aligned	Severe	0.429	0.430	0.433	0.461
BRACS	Reversed	Severe	0.382	0.398	0.385	0.473
BRACS	Native	Severe (spread)	0.432	0.447	0.451	0.503
BRACS	Aligned	Severe (spread)	0.431	0.427	0.425	0.478
BRACS	Reversed	Severe (spread)	0.375	0.378	0.379	0.491

Table 23: Equal-strength diagnostic for independent-support weighting, reported as balanced accuracy.

Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status	Discrimination R_M
BRACS	Aligned	Moderate	Prevalence	1.197	[0.999, 1.537]	<0.0001	0.0004	Tested	—
BRACS	Aligned	Moderate	Nominal support	1.227	[1.006, 1.708]	<0.0001	0.0004	Tested	—
BRACS	Aligned	Moderate	Independent support	-0.838	[-2.759, -0.105]	0.0002	0.0013	Tested	—
BRACS	Aligned	Moderate	Difficulty	-1.352	[-6.467, 0.395]	<0.0001	0.0004	Tested	—
BRACS	Aligned	Moderate	Diversity	-1.125	[-2.574, -0.481]	<0.0001	0.0004	Tested	—
BRACS	Reversed	Moderate	Prevalence	0.887	[0.441, 0.997]	<0.0001	0.0004	Tested	0.840
BRACS	Reversed	Moderate	Nominal support	0.873	[0.353, 0.977]	<0.0001	0.0004	Tested	0.833
BRACS	Reversed	Moderate	Independent support	-0.274	[-4.425, 0.494]	0.0033	0.0098	Tested	0.051
BRACS	Reversed	Moderate	Difficulty	-1.095	[-6.533, -0.014]	<0.0001	0.0004	Tested	-0.078
BRACS	Reversed	Moderate	Diversity	-1.303	[-8.308, 0.186]	<0.0001	0.0004	Tested	-0.064
BRACS	Native	Severe	Prevalence	0.911	[0.743, 1.008]	<0.0001	0.0004	Tested	0.768
BRACS	Native	Severe	Nominal support	0.682	[0.177, 0.885]	0.0018	0.0070	Tested	0.637
BRACS	Native	Severe	Independent support	-1.287	[-3.215, -0.212]	0.0003	0.0015	Tested	0.106
BRACS	Native	Severe	Difficulty	-2.168	[-5.847, -0.879]	<0.0001	0.0004	Tested	-0.273
BRACS	Native	Severe	Diversity	-1.753	[-4.845, -0.655]	<0.0001	0.0004	Tested	-0.165
BRACS	Aligned	Severe	Prevalence	1.137	[1.043, 1.261]	<0.0001	0.0004	Tested	0.654
BRACS	Aligned	Severe	Nominal support	0.539	[0.069, 0.913]	<0.0001	0.0004	Tested	0.426
BRACS	Aligned	Severe	Independent support	-2.509	[-3.840, -1.587]	<0.0001	0.0004	Tested	0.024
BRACS	Aligned	Severe	Difficulty	-1.768	[-3.203, -0.779]	<0.0001	0.0004	Tested	0.164
BRACS	Aligned	Severe	Diversity	-3.570	[-5.968, -2.334]	<0.0001	0.0004	Tested	-0.059
BRACS	Reversed	Severe	Prevalence	0.948	[0.896, 0.984]	<0.0001	0.0004	Tested	0.794
BRACS	Reversed	Severe	Nominal support	0.880	[0.807, 0.935]	<0.0001	0.0004	Tested	0.736
BRACS	Reversed	Severe	Independent support	-0.388	[-0.855, -0.137]	0.0002	0.0013	Tested	0.111
BRACS	Reversed	Severe	Difficulty	-1.158	[-1.812, -0.685]	<0.0001	0.0004	Tested	-0.059
BRACS	Reversed	Severe	Diversity	-1.060	[-1.951, -0.523]	<0.0001	0.0004	Tested	-0.010
BRACS	Native	Severe (spread)	Prevalence	-0.407	[-1.938, 0.217]	0.2420	0.4841	Tested	0.871
BRACS	Native	Severe (spread)	Nominal support	-0.442	[-2.019, 0.200]	0.5092	0.5092	Tested	0.831
BRACS	Native	Severe (spread)	Independent support	-3.233	[-7.753, -1.456]	<0.0001	0.0004	Tested	0.111
BRACS	Native	Severe (spread)	Difficulty	-4.058	[-8.884, -2.094]	<0.0001	0.0004	Tested	0.021
BRACS	Native	Severe (spread)	Diversity	-3.893	[-9.552, -1.655]	<0.0001	0.0004	Tested	-0.026
BRACS	Aligned	Severe (spread)	Prevalence	0.740	[0.576, 0.832]	<0.0001	0.0004	Tested	0.466
BRACS	Aligned	Severe (spread)	Nominal support	0.742	[0.582, 0.827]	<0.0001	0.0004	Tested	0.455
BRACS	Aligned	Severe (spread)	Independent support	-0.542	[-1.125, -0.168]	<0.0001	0.0004	Tested	-0.135
BRACS	Aligned	Severe (spread)	Difficulty	-0.345	[-0.717, -0.042]	<0.0001	0.0004	Tested	-0.145
BRACS	Aligned	Severe (spread)	Diversity	-0.897	[-1.878, -0.192]	<0.0001	0.0004	Tested	-0.145
BRACS	Reversed	Severe (spread)	Prevalence	0.719	[0.597, 0.806]	<0.0001	0.0004	Tested	0.824
BRACS	Reversed	Severe (spread)	Nominal support	0.663	[0.486, 0.776]	<0.0001	0.0004	Tested	0.791

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Dataset	Assignment	Severity	Signal	R_M	95% CI (bootstrap)	p (permutation)	Holm p	Status	Discrimination R_M
BRACS	Reversed	Severe (spread)	Independent support	-1.110	[-1.950, -0.603]	$<$0.0001	0.0004	Tested	0.051
BRACS	Reversed	Severe (spread)	Difficulty	-1.612	[-2.555, -0.991]	$<$0.0001	0.0004	Tested	-0.008
BRACS	Reversed	Severe (spread)	Diversity	-1.012	[-1.890, -0.480]	$<$0.0001	0.0004	Tested	0.030

Table 25: Confirmatory recovery of tail-group negative log likelihood by condition and weighting signal.

Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
BRACS	Native	Balanced	Discrimination	Balanced sampling	-0.211	[-0.906, 0.074]
BRACS	Native	Balanced	Discrimination	CE + soft F_1	-0.034	[-0.496, 0.221]
BRACS	Native	Balanced	Discrimination	CE + soft MCC	-0.499	[-1.801, -0.003]
BRACS	Native	Balanced	Discrimination	CFAL	-0.417	[-1.873, 0.171]
BRACS	Native	Balanced	Discrimination	cRT	-0.343	[-1.159, -0.038]
BRACS	Native	Balanced	Discrimination	Focal loss	-0.671	[-2.294, -0.132]
BRACS	Native	Balanced	Discrimination	Logit adjustment (train)	-0.005	[-0.034, 0.019]
BRACS	Native	Balanced	Discrimination	OKO	-0.953	[-3.348, -0.210]
BRACS	Native	Balanced	Discrimination	Logit adjustment (post-hoc)	-0.002	[-0.018, -0.000]
BRACS	Aligned	Balanced	Discrimination	Balanced sampling	-0.202	[-0.930, 0.068]
BRACS	Aligned	Balanced	Discrimination	CE + soft F_1	-0.033	[-0.508, 0.195]
BRACS	Aligned	Balanced	Discrimination	CE + soft MCC	-0.479	[-1.760, -0.001]
BRACS	Aligned	Balanced	Discrimination	CFAL	-0.400	[-1.842, 0.160]
BRACS	Aligned	Balanced	Discrimination	cRT	-0.329	[-1.165, -0.035]
BRACS	Aligned	Balanced	Discrimination	Focal loss	-0.644	[-2.303, -0.119]
BRACS	Aligned	Balanced	Discrimination	Logit adjustment (train)	-0.005	[-0.033, 0.018]
BRACS	Aligned	Balanced	Discrimination	OKO	-0.915	[-3.312, -0.198]
BRACS	Aligned	Balanced	Discrimination	Logit adjustment (post-hoc)	-0.002	[-0.017, -0.000]
BRACS	Reversed	Balanced	Discrimination	Balanced sampling	-0.234	[-1.373, 0.090]
BRACS	Reversed	Balanced	Discrimination	CE + soft F_1	-0.038	[-0.673, 0.277]
BRACS	Reversed	Balanced	Discrimination	CE + soft MCC	-0.555	[-2.738, 0.018]
BRACS	Reversed	Balanced	Discrimination	CFAL	-0.464	[-2.730, 0.225]
BRACS	Reversed	Balanced	Discrimination	cRT	-0.381	[-1.834, -0.029]
BRACS	Reversed	Balanced	Discrimination	Focal loss	-0.746	[-3.529, -0.122]
BRACS	Reversed	Balanced	Discrimination	Logit adjustment (train)	-0.006	[-0.052, 0.024]
BRACS	Reversed	Balanced	Discrimination	OKO	-1.060	[-5.348, -0.202]
BRACS	Reversed	Balanced	Discrimination	Logit adjustment (post-hoc)	-0.003	[-0.027, -0.000]
BRACS	Native	Moderate	Discrimination	Balanced sampling	0.715	[0.305, 1.033]
BRACS	Native	Moderate	Discrimination	CE + soft F_1	0.714	[0.303, 1.038]
BRACS	Native	Moderate	Discrimination	CE + soft MCC	0.714	[0.303, 1.038]
BRACS	Native	Moderate	Discrimination	CFAL	0.501	[0.150, 0.847]
BRACS	Native	Moderate	Discrimination	cRT	0.381	[-0.132, 0.766]
BRACS	Native	Moderate	Discrimination	Focal loss	0.770	[0.412, 1.078]
BRACS	Native	Moderate	Discrimination	Logit adjustment (train)	0.866	[0.553, 1.237]
BRACS	Native	Moderate	Discrimination	OKO	0.462	[-0.190, 0.988]
BRACS	Native	Moderate	Discrimination	Logit adjustment (post-hoc)	0.234	[0.040, 0.435]
BRACS	Reversed	Moderate	Discrimination	Balanced sampling	0.796	[0.661, 0.930]
BRACS	Reversed	Moderate	Discrimination	CE + soft F_1	0.795	[0.660, 0.930]
BRACS	Reversed	Moderate	Discrimination	CE + soft MCC	0.795	[0.660, 0.930]
BRACS	Reversed	Moderate	Discrimination	CFAL	0.311	[0.112, 0.517]
BRACS	Reversed	Moderate	Discrimination	cRT	0.192	[0.010, 0.354]
BRACS	Reversed	Moderate	Discrimination	Focal loss	0.828	[0.705, 0.957]
BRACS	Reversed	Moderate	Discrimination	Logit adjustment (train)	0.800	[0.658, 0.932]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
BRACS	Reversed	Moderate	Discrimination	OKO	0.742	[0.511, 0.940]
BRACS	Reversed	Moderate	Discrimination	Logit adjustment (post-hoc)	0.233	[0.145, 0.337]
BRACS	Native	Severe	Discrimination	Balanced sampling	0.797	[0.467, 1.067]
BRACS	Native	Severe	Discrimination	CE + soft F_1	0.669	[0.370, 0.901]
BRACS	Native	Severe	Discrimination	CE + soft MCC	0.798	[0.467, 1.069]
BRACS	Native	Severe	Discrimination	CFAL	0.587	[0.357, 0.839]
BRACS	Native	Severe	Discrimination	cRT	0.230	[-0.082, 0.465]
BRACS	Native	Severe	Discrimination	Focal loss	0.725	[0.352, 0.986]
BRACS	Native	Severe	Discrimination	Logit adjustment (train)	0.689	[0.383, 0.948]
BRACS	Native	Severe	Discrimination	OKO	0.518	[0.178, 0.801]
BRACS	Native	Severe	Discrimination	Logit adjustment (post-hoc)	0.385	[0.252, 0.546]
BRACS	Aligned	Severe	Discrimination	Balanced sampling	0.731	[0.408, 1.135]
BRACS	Aligned	Severe	Discrimination	CE + soft F_1	0.674	[0.347, 1.081]
BRACS	Aligned	Severe	Discrimination	CE + soft MCC	0.731	[0.409, 1.135]
BRACS	Aligned	Severe	Discrimination	CFAL	0.383	[0.008, 0.771]
BRACS	Aligned	Severe	Discrimination	cRT	0.052	[-0.351, 0.337]
BRACS	Aligned	Severe	Discrimination	Focal loss	0.598	[0.225, 0.956]
BRACS	Aligned	Severe	Discrimination	Logit adjustment (train)	0.217	[-0.305, 0.554]
BRACS	Aligned	Severe	Discrimination	OKO	0.792	[0.465, 1.274]
BRACS	Aligned	Severe	Discrimination	Logit adjustment (post-hoc)	0.007	[-0.328, 0.216]
BRACS	Reversed	Severe	Discrimination	Balanced sampling	0.777	[0.664, 0.874]
BRACS	Reversed	Severe	Discrimination	CE + soft F_1	0.790	[0.680, 0.894]
BRACS	Reversed	Severe	Discrimination	CE + soft MCC	0.776	[0.662, 0.873]
BRACS	Reversed	Severe	Discrimination	CFAL	0.494	[0.354, 0.628]
BRACS	Reversed	Severe	Discrimination	cRT	0.438	[0.330, 0.565]
BRACS	Reversed	Severe	Discrimination	Focal loss	0.807	[0.699, 0.905]
BRACS	Reversed	Severe	Discrimination	Logit adjustment (train)	0.717	[0.616, 0.827]
BRACS	Reversed	Severe	Discrimination	OKO	0.785	[0.646, 0.929]
BRACS	Reversed	Severe	Discrimination	Logit adjustment (post-hoc)	0.179	[0.124, 0.244]
BRACS	Native	Severe (spread)	Discrimination	Balanced sampling	0.908	[0.765, 1.086]
BRACS	Native	Severe (spread)	Discrimination	CE + soft F_1	0.908	[0.765, 1.086]
BRACS	Native	Severe (spread)	Discrimination	CE + soft MCC	0.908	[0.765, 1.086]
BRACS	Native	Severe (spread)	Discrimination	CFAL	0.487	[0.324, 0.667]
BRACS	Native	Severe (spread)	Discrimination	cRT	0.385	[0.212, 0.548]
BRACS	Native	Severe (spread)	Discrimination	Focal loss	0.863	[0.667, 1.084]
BRACS	Native	Severe (spread)	Discrimination	Logit adjustment (train)	0.723	[0.520, 0.919]
BRACS	Native	Severe (spread)	Discrimination	OKO	0.720	[0.532, 0.921]
BRACS	Native	Severe (spread)	Discrimination	Logit adjustment (post-hoc)	0.615	[0.408, 0.823]
BRACS	Aligned	Severe (spread)	Discrimination	Balanced sampling	0.436	[-0.009, 0.735]
BRACS	Aligned	Severe (spread)	Discrimination	CE + soft F_1	0.436	[-0.009, 0.735]
BRACS	Aligned	Severe (spread)	Discrimination	CE + soft MCC	0.436	[-0.009, 0.735]
BRACS	Aligned	Severe (spread)	Discrimination	CFAL	0.169	[-0.181, 0.498]
BRACS	Aligned	Severe (spread)	Discrimination	cRT	-0.260	[-1.016, 0.068]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
BRACS	Aligned	Severe (spread)	Discrimination	Focal loss	0.377	[-0.187, 0.741]
BRACS	Aligned	Severe (spread)	Discrimination	Logit adjustment (train)	0.124	[-0.576, 0.514]
BRACS	Aligned	Severe (spread)	Discrimination	OKO	0.554	[0.232, 0.927]
BRACS	Aligned	Severe (spread)	Discrimination	Logit adjustment (post-hoc)	-0.393	[-1.419, 0.045]
BRACS	Reversed	Severe (spread)	Discrimination	Balanced sampling	0.854	[0.753, 0.957]
BRACS	Reversed	Severe (spread)	Discrimination	CE + soft F_1	0.854	[0.753, 0.957]
BRACS	Reversed	Severe (spread)	Discrimination	CE + soft MCC	0.854	[0.753, 0.957]
BRACS	Reversed	Severe (spread)	Discrimination	CFAL	0.455	[0.346, 0.578]
BRACS	Reversed	Severe (spread)	Discrimination	cRT	0.427	[0.340, 0.521]
BRACS	Reversed	Severe (spread)	Discrimination	Focal loss	0.831	[0.725, 0.942]
BRACS	Reversed	Severe (spread)	Discrimination	Logit adjustment (train)	0.835	[0.725, 0.954]
BRACS	Reversed	Severe (spread)	Discrimination	OKO	0.834	[0.713, 0.943]
BRACS	Reversed	Severe (spread)	Discrimination	Logit adjustment (post-hoc)	0.462	[0.376, 0.557]
BRACS	Aligned	Moderate	Calibration	Balanced sampling	1.355	[1.096, 2.128]
BRACS	Aligned	Moderate	Calibration	CE + soft F_1	1.355	[1.097, 2.129]
BRACS	Aligned	Moderate	Calibration	CE + soft MCC	1.355	[1.097, 2.129]
BRACS	Aligned	Moderate	Calibration	CFAL	0.529	[-0.212, 0.900]
BRACS	Aligned	Moderate	Calibration	cRT	-1.278	[-4.556, -0.299]
BRACS	Aligned	Moderate	Calibration	Focal loss	1.356	[1.165, 1.900]
BRACS	Aligned	Moderate	Calibration	Logit adjustment (train)	0.745	[0.516, 0.952]
BRACS	Aligned	Moderate	Calibration	OKO	1.165	[0.948, 1.644]
BRACS	Aligned	Moderate	Calibration	Logit adjustment (post-hoc)	-0.874	[-2.066, -0.485]
BRACS	Reversed	Moderate	Calibration	Balanced sampling	0.899	[0.581, 0.981]
BRACS	Reversed	Moderate	Calibration	CE + soft F_1	0.898	[0.582, 0.981]
BRACS	Reversed	Moderate	Calibration	CE + soft MCC	0.898	[0.582, 0.981]
BRACS	Reversed	Moderate	Calibration	CFAL	0.305	[-2.220, 0.735]
BRACS	Reversed	Moderate	Calibration	cRT	-0.461	[-4.132, 0.325]
BRACS	Reversed	Moderate	Calibration	Focal loss	0.898	[0.544, 0.999]
BRACS	Reversed	Moderate	Calibration	Logit adjustment (train)	1.040	[0.944, 1.383]
BRACS	Reversed	Moderate	Calibration	OKO	0.692	[-0.294, 0.878]
BRACS	Reversed	Moderate	Calibration	Logit adjustment (post-hoc)	-0.315	[-1.241, -0.144]
BRACS	Native	Severe	Calibration	Balanced sampling	0.942	[0.805, 1.043]
BRACS	Native	Severe	Calibration	CE + soft F_1	0.838	[0.649, 0.954]
BRACS	Native	Severe	Calibration	CE + soft MCC	0.942	[0.805, 1.043]
BRACS	Native	Severe	Calibration	CFAL	0.474	[-0.201, 0.726]
BRACS	Native	Severe	Calibration	cRT	-0.425	[-1.435, 0.063]
BRACS	Native	Severe	Calibration	Focal loss	0.895	[0.740, 1.001]
BRACS	Native	Severe	Calibration	Logit adjustment (train)	0.980	[0.742, 1.092]
BRACS	Native	Severe	Calibration	OKO	0.813	[0.494, 0.975]
BRACS	Native	Severe	Calibration	Logit adjustment (post-hoc)	-0.541	[-1.094, -0.340]
BRACS	Aligned	Severe	Calibration	Balanced sampling	1.198	[1.093, 1.360]
BRACS	Aligned	Severe	Calibration	CE + soft F_1	1.152	[1.025, 1.283]
BRACS	Aligned	Severe	Calibration	CE + soft MCC	1.198	[1.093, 1.360]

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Dataset	Assignment	Severity	Gate	Method	R_M	95% CI (bootstrap)
BRACS	Aligned	Severe	Calibration	CFAL	0.949	[0.763, 1.089]
BRACS	Aligned	Severe	Calibration	cRT	-1.790	[-3.839, -0.771]
BRACS	Aligned	Severe	Calibration	Focal loss	1.213	[1.118, 1.358]
BRACS	Aligned	Severe	Calibration	Logit adjustment (train)	0.620	[0.329, 0.781]
BRACS	Aligned	Severe	Calibration	OKO	1.065	[0.952, 1.190]
BRACS	Aligned	Severe	Calibration	Logit adjustment (post-hoc)	-0.917	[-1.382, -0.637]
BRACS	Reversed	Severe	Calibration	Balanced sampling	0.922	[0.862, 0.959]
BRACS	Reversed	Severe	Calibration	CE + soft F_1	0.931	[0.878, 0.970]
BRACS	Reversed	Severe	Calibration	CE + soft MCC	0.920	[0.861, 0.958]
BRACS	Reversed	Severe	Calibration	CFAL	0.646	[0.448, 0.752]
BRACS	Reversed	Severe	Calibration	cRT	0.177	[-0.218, 0.445]
BRACS	Reversed	Severe	Calibration	Focal loss	0.961	[0.905, 1.003]
BRACS	Reversed	Severe	Calibration	Logit adjustment (train)	0.998	[0.949, 1.041]
BRACS	Reversed	Severe	Calibration	OKO	0.861	[0.766, 0.918]
BRACS	Reversed	Severe	Calibration	Logit adjustment (post-hoc)	-0.357	[-0.484, -0.287]
BRACS	Native	Severe (spread)	Calibration	Balanced sampling	-0.406	[-2.034, 0.223]
BRACS	Native	Severe (spread)	Calibration	CE + soft F_1	-0.406	[-2.034, 0.223]
BRACS	Native	Severe (spread)	Calibration	CE + soft MCC	-0.406	[-2.034, 0.223]
BRACS	Native	Severe (spread)	Calibration	CFAL	-2.366	[-6.130, -0.914]
BRACS	Native	Severe (spread)	Calibration	cRT	-3.681	[-7.745, -2.009]
BRACS	Native	Severe (spread)	Calibration	Focal loss	-0.370	[-2.057, 0.272]
BRACS	Native	Severe (spread)	Calibration	Logit adjustment (train)	0.822	[0.580, 1.117]
BRACS	Native	Severe (spread)	Calibration	OKO	-0.673	[-2.569, 0.077]
BRACS	Native	Severe (spread)	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
BRACS	Aligned	Severe (spread)	Calibration	Balanced sampling	0.713	[0.522, 0.817]
BRACS	Aligned	Severe (spread)	Calibration	CE + soft F_1	0.713	[0.522, 0.817]
BRACS	Aligned	Severe (spread)	Calibration	CE + soft MCC	0.713	[0.522, 0.817]
BRACS	Aligned	Severe (spread)	Calibration	CFAL	0.260	[-0.235, 0.515]
BRACS	Aligned	Severe (spread)	Calibration	cRT	-0.641	[-1.401, -0.241]
BRACS	Aligned	Severe (spread)	Calibration	Focal loss	0.731	[0.561, 0.826]
BRACS	Aligned	Severe (spread)	Calibration	Logit adjustment (train)	0.915	[0.855, 0.969]
BRACS	Aligned	Severe (spread)	Calibration	OKO	0.725	[0.570, 0.812]
BRACS	Aligned	Severe (spread)	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]
BRACS	Reversed	Severe (spread)	Calibration	Balanced sampling	0.715	[0.596, 0.796]
BRACS	Reversed	Severe (spread)	Calibration	CE + soft F_1	0.715	[0.596, 0.796]
BRACS	Reversed	Severe (spread)	Calibration	CE + soft MCC	0.715	[0.596, 0.796]
BRACS	Reversed	Severe (spread)	Calibration	CFAL	0.342	[0.051, 0.536]
BRACS	Reversed	Severe (spread)	Calibration	cRT	-0.447	[-0.900, -0.190]
BRACS	Reversed	Severe (spread)	Calibration	Focal loss	0.731	[0.612, 0.817]
BRACS	Reversed	Severe (spread)	Calibration	Logit adjustment (train)	0.989	[0.944, 1.029]
BRACS	Reversed	Severe (spread)	Calibration	OKO	0.708	[0.566, 0.804]
BRACS	Reversed	Severe (spread)	Calibration	Logit adjustment (post-hoc)	0.000	[0.000, 0.000]

Table 27: Exploratory recovery estimates for the wider method roster.

D Per-Class and Tier Endpoints

Section 2.4 reports damage at tier level; this appendix gives the per-class detail behind it. Recall, F_1, negative log likelihood, and Brier score are listed per class and condition for the cross-entropy reference, averaged within a partition and then with equal weight across the three partitions, with the tier assignment given alongside. The table is restricted to cross-entropy because that is the reference against which every deficit in Section 2 is defined. As recorded in Appendix A.2, the per-partition cells for the ADH, DCIS, and FEA classes are designated descriptive-only by the preflight, so their entries here carry no inferential weight.

Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
BRACS	Aligned	Balanced (spread)	ADH	body	0.055	0.062	2.440	1.076
BRACS	Aligned	Balanced (spread)	ADH	tail	0.210	0.216	2.825	1.217
BRACS	Aligned	Balanced (spread)	DCIS	body	0.480	0.504	0.804	0.365
BRACS	Aligned	Balanced (spread)	DCIS	head	0.476	0.368	1.038	0.541
BRACS	Aligned	Balanced (spread)	DCIS	tail	0.634	0.652	1.205	0.631
BRACS	Aligned	Balanced (spread)	FEA	body	0.439	0.307	3.524	1.346
BRACS	Aligned	Balanced (spread)	FEA	head	0.446	0.377	1.885	0.846
BRACS	Aligned	Balanced (spread)	IC	head	0.880	0.904	0.398	0.168
BRACS	Aligned	Balanced (spread)	N	head	0.673	0.473	1.065	0.571
BRACS	Aligned	Balanced (spread)	PB	tail	0.460	0.582	0.877	0.480
BRACS	Aligned	Balanced (spread)	UDH	tail	0.414	0.343	1.919	0.933
BRACS	Aligned	Moderate	ADH	body	0.186	0.121	3.311	1.200
BRACS	Aligned	Moderate	ADH	tail	0.037	0.053	3.249	1.313
BRACS	Aligned	Moderate	DCIS	body	0.557	0.490	0.907	0.385
BRACS	Aligned	Moderate	DCIS	head	0.614	0.411	1.242	0.490
BRACS	Aligned	Moderate	DCIS	tail	0.403	0.523	1.137	0.519
BRACS	Aligned	Moderate	FEA	body	0.385	0.195	4.278	1.429
BRACS	Aligned	Moderate	FEA	head	0.545	0.268	2.371	0.898
BRACS	Aligned	Moderate	IC	head	0.883	0.892	0.646	0.234
BRACS	Aligned	Moderate	N	head	0.840	0.431	1.111	0.576
BRACS	Aligned	Moderate	PB	tail	0.186	0.293	1.496	0.732
BRACS	Aligned	Moderate	UDH	tail	0.206	0.242	2.063	0.951
BRACS	Aligned	Severe	ADH	body	0.246	0.136	5.255	1.306
BRACS	Aligned	Severe	ADH	tail	0.020	0.034	3.521	1.316
BRACS	Aligned	Severe	DCIS	body	0.514	0.472	1.534	0.571
BRACS	Aligned	Severe	DCIS	head	0.568	0.410	1.192	0.504
BRACS	Aligned	Severe	DCIS	tail	0.241	0.356	1.743	0.702
BRACS	Aligned	Severe	FEA	body	0.469	0.180	3.554	1.289
BRACS	Aligned	Severe	FEA	head	0.497	0.235	3.545	1.030
BRACS	Aligned	Severe	IC	head	0.889	0.882	0.919	0.266
BRACS	Aligned	Severe	N	head	0.832	0.406	1.485	0.684
BRACS	Aligned	Severe	PB	tail	0.133	0.222	1.913	0.844
BRACS	Aligned	Severe	UDH	tail	0.129	0.154	2.407	0.995
BRACS	Aligned	Severe (spread)	ADH	body	0.308	0.196	3.679	1.228
BRACS	Aligned	Severe (spread)	ADH	tail	0.021	0.038	5.225	1.440
BRACS	Aligned	Severe (spread)	DCIS	body	0.511	0.464	1.448	0.529
BRACS	Aligned	Severe (spread)	DCIS	head	0.390	0.258	2.465	0.806
BRACS	Aligned	Severe (spread)	DCIS	tail	0.197	0.310	1.493	0.640
BRACS	Aligned	Severe (spread)	FEA	body	0.406	0.195	5.132	1.317
BRACS	Aligned	Severe (spread)	FEA	head	0.478	0.249	3.601	1.036
BRACS	Aligned	Severe (spread)	IC	head	0.904	0.886	0.824	0.234
BRACS	Aligned	Severe (spread)	N	head	0.862	0.435	1.273	0.571
BRACS	Aligned	Severe (spread)	PB	tail	0.184	0.287	1.688	0.759

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Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
BRACS	Aligned	Severe (spread)	UDH	tail	0.133	0.167	2.884	1.101
BRACS	Reversed	Balanced (spread)	ADH	body	0.068	0.070	2.388	1.062
BRACS	Reversed	Balanced (spread)	ADH	head	0.230	0.220	2.768	1.195
BRACS	Reversed	Balanced (spread)	DCIS	body	0.449	0.485	0.819	0.375
BRACS	Reversed	Balanced (spread)	DCIS	head	0.606	0.638	1.226	0.643
BRACS	Reversed	Balanced (spread)	DCIS	tail	0.504	0.382	0.990	0.512
BRACS	Reversed	Balanced (spread)	FEA	body	0.419	0.299	3.561	1.333
BRACS	Reversed	Balanced (spread)	FEA	tail	0.438	0.369	1.896	0.849
BRACS	Reversed	Balanced (spread)	IC	tail	0.876	0.903	0.419	0.174
BRACS	Reversed	Balanced (spread)	N	tail	0.672	0.466	1.076	0.576
BRACS	Reversed	Balanced (spread)	PB	head	0.441	0.566	0.887	0.485
BRACS	Reversed	Balanced (spread)	UDH	head	0.414	0.334	1.901	0.924
BRACS	Reversed	Moderate	ADH	body	0.063	0.063	2.708	1.080
BRACS	Reversed	Moderate	ADH	head	0.441	0.180	3.601	1.308
BRACS	Reversed	Moderate	DCIS	body	0.329	0.401	1.816	0.606
BRACS	Reversed	Moderate	DCIS	head	0.832	0.457	1.255	0.628
BRACS	Reversed	Moderate	DCIS	tail	0.364	0.347	1.871	0.686
BRACS	Reversed	Moderate	FEA	body	0.198	0.169	6.328	1.495
BRACS	Reversed	Moderate	FEA	tail	0.305	0.278	3.072	0.904
BRACS	Reversed	Moderate	IC	tail	0.743	0.825	0.835	0.229
BRACS	Reversed	Moderate	N	tail	0.311	0.329	1.583	0.683
BRACS	Reversed	Moderate	PB	head	0.348	0.449	2.411	0.902
BRACS	Reversed	Moderate	UDH	head	0.464	0.296	2.643	0.995
BRACS	Reversed	Severe	ADH	body	0.094	0.090	5.539	1.247
BRACS	Reversed	Severe	ADH	head	0.553	0.173	4.036	1.281
BRACS	Reversed	Severe	DCIS	body	0.240	0.300	2.604	0.740
BRACS	Reversed	Severe	DCIS	head	0.826	0.337	1.675	0.564
BRACS	Reversed	Severe	DCIS	tail	0.313	0.334	2.878	0.856
BRACS	Reversed	Severe	FEA	body	0.126	0.103	8.673	1.575
BRACS	Reversed	Severe	FEA	tail	0.201	0.214	4.261	1.076
BRACS	Reversed	Severe	IC	tail	0.605	0.732	1.182	0.348
BRACS	Reversed	Severe	N	tail	0.222	0.270	2.518	0.865
BRACS	Reversed	Severe	PB	head	0.364	0.463	3.632	1.022
BRACS	Reversed	Severe	UDH	head	0.447	0.273	3.589	1.095
BRACS	Reversed	Severe (spread)	ADH	body	0.051	0.053	4.132	1.247
BRACS	Reversed	Severe (spread)	ADH	head	0.583	0.178	3.659	1.191
BRACS	Reversed	Severe (spread)	DCIS	body	0.145	0.195	1.704	0.645
BRACS	Reversed	Severe (spread)	DCIS	head	0.823	0.342	1.540	0.591
BRACS	Reversed	Severe (spread)	DCIS	tail	0.195	0.192	3.120	0.952
BRACS	Reversed	Severe (spread)	FEA	body	0.204	0.164	6.270	1.461
BRACS	Reversed	Severe (spread)	FEA	tail	0.192	0.204	3.394	1.048
BRACS	Reversed	Severe (spread)	IC	tail	0.562	0.706	0.984	0.345
BRACS	Reversed	Severe (spread)	N	tail	0.202	0.266	1.836	0.742

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Dataset	Assignment	Condition	Class	Tier	Recall	F_1	NLL	Brier
BRACS	Reversed	Severe (spread)	PB	head	0.465	0.538	2.277	0.855
BRACS	Reversed	Severe (spread)	UDH	head	0.409	0.263	3.562	1.143
BRACS	Native	Balanced (spread)	ADH	tail	0.168	0.170	2.668	1.160
BRACS	Native	Balanced (spread)	DCIS	tail	0.537	0.508	1.008	0.507
BRACS	Native	Balanced (spread)	FEA	body	0.443	0.350	2.440	1.011
BRACS	Native	Balanced (spread)	IC	tail	0.872	0.901	0.433	0.179
BRACS	Native	Balanced (spread)	N	head	0.673	0.467	1.059	0.565
BRACS	Native	Balanced (spread)	PB	head	0.466	0.585	0.876	0.481
BRACS	Native	Balanced (spread)	UDH	head	0.404	0.345	1.931	0.936
BRACS	Native	Moderate	ADH	tail	0.082	0.089	2.972	1.246
BRACS	Native	Moderate	DCIS	tail	0.443	0.461	1.098	0.550
BRACS	Native	Moderate	FEA	body	0.443	0.310	2.739	1.031
BRACS	Native	Moderate	IC	tail	0.826	0.871	0.524	0.199
BRACS	Native	Moderate	N	head	0.731	0.428	1.248	0.644
BRACS	Native	Moderate	PB	head	0.319	0.430	1.706	0.768
BRACS	Native	Moderate	UDH	head	0.426	0.326	2.054	0.914
BRACS	Native	Severe	ADH	tail	0.095	0.104	3.842	1.315
BRACS	Native	Severe	DCIS	tail	0.395	0.442	1.424	0.591
BRACS	Native	Severe	FEA	body	0.417	0.316	3.337	1.104
BRACS	Native	Severe	IC	tail	0.742	0.824	0.663	0.234
BRACS	Native	Severe	N	head	0.764	0.409	1.341	0.589
BRACS	Native	Severe	PB	head	0.293	0.390	2.779	0.967
BRACS	Native	Severe	UDH	head	0.421	0.306	2.470	0.928
BRACS	Native	Severe (spread)	ADH	tail	0.118	0.119	3.014	1.201
BRACS	Native	Severe (spread)	DCIS	tail	0.259	0.309	1.281	0.636
BRACS	Native	Severe (spread)	FEA	body	0.376	0.331	3.003	1.110
BRACS	Native	Severe (spread)	IC	tail	0.719	0.817	0.402	0.184
BRACS	Native	Severe (spread)	N	head	0.748	0.421	1.369	0.617
BRACS	Native	Severe (spread)	PB	head	0.401	0.469	1.425	0.719
BRACS	Native	Severe (spread)	UDH	head	0.403	0.298	2.369	0.990

Table 29: Equal-weight three-partition per-class test endpoints for the cross-entropy reference by condition, with the tier assignment each condition induces.

E Calibration Detail

The calibration gate reads tail-group macro negative log likelihood, but the shape of the miscalibration matters for method design and is not visible in a scalar. This appendix reports the raw and temperature-scaled expected calibration error, negative log likelihood, and Brier score per condition and method for the cross-entropy reference and the confirmatory family, and shows the reliability curves from which they are computed.

Dataset	Assignment	Condition	Method	NLL	Brier	ECE	Temp. NLL	Temp. Brier	Temp. ECE
BRACS	Aligned	Balanced (spread)	CE	0.917	0.453	0.041 [0.031, 0.079]	0.919	0.453	0.047
BRACS	Aligned	Balanced (spread)	Nominal support	0.950	0.471	0.041 [0.032, 0.073]	0.945	0.470	0.038
BRACS	Aligned	Balanced (spread)	Independent support	0.952	0.472	0.040 [0.032, 0.072]	0.948	0.471	0.037
BRACS	Aligned	Balanced (spread)	Difficulty	0.955	0.471	0.042 [0.033, 0.073]	0.949	0.470	0.037
BRACS	Aligned	Balanced (spread)	Diversity	0.955	0.473	0.042 [0.032, 0.074]	0.950	0.472	0.038
BRACS	Aligned	Balanced (spread)	Prevalence	0.951	0.471	0.041 [0.032, 0.073]	0.946	0.470	0.038
BRACS	Aligned	Moderate	CE	1.211	0.548	0.143 [0.108, 0.189]	1.047	0.510	0.059
BRACS	Aligned	Moderate	Nominal support	1.099	0.534	0.083 [0.060, 0.130]	1.066	0.524	0.062
BRACS	Aligned	Moderate	Independent support	1.223	0.578	0.170 [0.122, 0.230]	1.087	0.529	0.072
BRACS	Aligned	Moderate	Difficulty	1.426	0.601	0.185 [0.130, 0.257]	1.119	0.545	0.088
BRACS	Aligned	Moderate	Diversity	1.385	0.614	0.205 [0.151, 0.276]	1.125	0.546	0.082
BRACS	Aligned	Moderate	Prevalence	1.130	0.543	0.091 [0.067, 0.139]	1.080	0.529	0.064
BRACS	Aligned	Severe	CE	1.551	0.621	0.219 [0.174, 0.280]	1.098	0.531	0.065
BRACS	Aligned	Severe	Nominal support	1.210	0.572	0.150 [0.105, 0.211]	1.098	0.535	0.075
BRACS	Aligned	Severe	Independent support	2.165	0.744	0.320 [0.249, 0.408]	1.191	0.567	0.099
BRACS	Aligned	Severe	Difficulty	1.930	0.736	0.312 [0.243, 0.401]	1.199	0.575	0.110
BRACS	Aligned	Severe	Diversity	2.617	0.784	0.349 [0.279, 0.438]	1.225	0.577	0.098
BRACS	Aligned	Severe	Prevalence	1.089	0.531	0.081 [0.060, 0.130]	1.063	0.526	0.073
BRACS	Aligned	Severe (spread)	CE	1.564	0.593	0.209 [0.160, 0.275]	1.064	0.514	0.064
BRACS	Aligned	Severe (spread)	Nominal support	1.000	0.498	0.072 [0.046, 0.116]	0.996	0.497	0.072
BRACS	Aligned	Severe (spread)	Independent support	2.304	0.749	0.328 [0.258, 0.418]	1.179	0.570	0.110
BRACS	Aligned	Severe (spread)	Difficulty	2.114	0.724	0.310 [0.241, 0.400]	1.162	0.565	0.105
BRACS	Aligned	Severe (spread)	Diversity	2.454	0.742	0.327 [0.253, 0.420]	1.165	0.561	0.101
BRACS	Aligned	Severe (spread)	Prevalence	0.971	0.484	0.055 [0.039, 0.094]	0.973	0.486	0.060
BRACS	Reversed	Balanced (spread)	CE	0.920	0.453	0.041 [0.030, 0.077]	0.924	0.454	0.046
BRACS	Reversed	Balanced (spread)	Nominal support	0.960	0.475	0.044 [0.033, 0.077]	0.953	0.474	0.039
BRACS	Reversed	Balanced (spread)	Independent support	0.964	0.476	0.048 [0.036, 0.081]	0.954	0.474	0.040
BRACS	Reversed	Balanced (spread)	Difficulty	0.982	0.483	0.055 [0.041, 0.087]	0.970	0.480	0.044
BRACS	Reversed	Balanced (spread)	Diversity	0.964	0.476	0.048 [0.036, 0.079]	0.955	0.475	0.040
BRACS	Reversed	Balanced (spread)	Prevalence	0.964	0.476	0.047 [0.035, 0.080]	0.956	0.475	0.040
BRACS	Reversed	Moderate	CE	1.769	0.635	0.215 [0.170, 0.276]	1.164	0.548	0.062
BRACS	Reversed	Moderate	Nominal support	1.087	0.527	0.068 [0.056, 0.104]	1.065	0.518	0.045
BRACS	Reversed	Moderate	Independent support	1.585	0.642	0.188 [0.154, 0.235]	1.224	0.579	0.081
BRACS	Reversed	Moderate	Difficulty	2.287	0.749	0.268 [0.227, 0.319]	1.322	0.624	0.088
BRACS	Reversed	Moderate	Diversity	2.301	0.743	0.270 [0.220, 0.327]	1.306	0.615	0.074
BRACS	Reversed	Moderate	Prevalence	1.095	0.529	0.067 [0.047, 0.109]	1.067	0.520	0.049
BRACS	Reversed	Severe	CE	2.582	0.751	0.313 [0.256, 0.384]	1.271	0.590	0.067
BRACS	Reversed	Severe	Nominal support	1.130	0.545	0.086 [0.065, 0.126]	1.104	0.534	0.062
BRACS	Reversed	Severe	Independent support	2.513	0.801	0.307 [0.261, 0.362]	1.364	0.651	0.093
BRACS	Reversed	Severe	Difficulty	3.392	0.920	0.401 [0.345, 0.463]	1.453	0.693	0.085
BRACS	Reversed	Severe	Diversity	3.102	0.857	0.352 [0.295, 0.415]	1.408	0.666	0.079
BRACS	Reversed	Severe	Prevalence	1.064	0.518	0.064 [0.050, 0.101]	1.052	0.514	0.055
BRACS	Reversed	Severe (spread)	CE	1.915	0.680	0.259 [0.208, 0.327]	1.180	0.563	0.048

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Dataset	Assignment	Condition	Method	NLL	Brier	ECE	Temp. NLL	Temp. Brier	Temp. ECE
BRACS	Reversed	Severe (spread)	Nominal support	1.037	0.509	0.074 [0.049, 0.108]	1.028	0.505	0.067
BRACS	Reversed	Severe (spread)	Independent support	2.653	0.864	0.358 [0.310, 0.414]	1.391	0.676	0.037
BRACS	Reversed	Severe (spread)	Difficulty	3.073	0.889	0.381 [0.332, 0.437]	1.405	0.682	0.049
BRACS	Reversed	Severe (spread)	Diversity	2.635	0.856	0.354 [0.306, 0.410]	1.391	0.674	0.037
BRACS	Reversed	Severe (spread)	Prevalence	1.007	0.494	0.061 [0.043, 0.091]	1.003	0.495	0.066
BRACS	Native	Balanced (spread)	CE	0.924	0.455	0.050 [0.036, 0.084]	0.922	0.454	0.048
BRACS	Native	Balanced (spread)	Nominal support	0.960	0.474	0.055 [0.040, 0.084]	0.951	0.472	0.044
BRACS	Native	Balanced (spread)	Independent support	0.960	0.474	0.055 [0.040, 0.085]	0.950	0.471	0.042
BRACS	Native	Balanced (spread)	Difficulty	0.967	0.477	0.053 [0.038, 0.082]	0.959	0.475	0.042
BRACS	Native	Balanced (spread)	Diversity	0.965	0.477	0.055 [0.042, 0.086]	0.954	0.474	0.043
BRACS	Native	Balanced (spread)	Prevalence	0.963	0.476	0.054 [0.040, 0.084]	0.954	0.474	0.045
BRACS	Native	Moderate	CE	1.274	0.564	0.148 [0.103, 0.210]	1.111	0.527	0.077
BRACS	Native	Moderate	Nominal support	1.023	0.504	0.061 [0.050, 0.092]	1.018	0.502	0.051
BRACS	Native	Moderate	Independent support	1.185	0.549	0.117 [0.080, 0.185]	1.091	0.525	0.067
BRACS	Native	Moderate	Difficulty	1.267	0.584	0.128 [0.085, 0.194]	1.150	0.555	0.069
BRACS	Native	Moderate	Diversity	1.355	0.592	0.137 [0.091, 0.205]	1.153	0.555	0.058
BRACS	Native	Moderate	Prevalence	1.031	0.506	0.058 [0.048, 0.095]	1.027	0.505	0.053
BRACS	Native	Severe	CE	1.793	0.638	0.220 [0.176, 0.276]	1.199	0.554	0.079
BRACS	Native	Severe	Nominal support	1.131	0.545	0.100 [0.067, 0.150]	1.089	0.530	0.062
BRACS	Native	Severe	Independent support	1.751	0.665	0.208 [0.143, 0.297]	1.259	0.599	0.084
BRACS	Native	Severe	Difficulty	2.330	0.734	0.271 [0.215, 0.343]	1.310	0.620	0.090
BRACS	Native	Severe	Diversity	2.085	0.714	0.247 [0.192, 0.320]	1.311	0.616	0.077
BRACS	Native	Severe	Prevalence	1.072	0.522	0.078 [0.053, 0.122]	1.051	0.517	0.060
BRACS	Native	Severe (spread)	CE	1.203	0.549	0.147 [0.112, 0.198]	1.062	0.514	0.062
BRACS	Native	Severe (spread)	Nominal support	0.979	0.487	0.054 [0.044, 0.086]	0.971	0.485	0.045
BRACS	Native	Severe (spread)	Independent support	1.369	0.629	0.161 [0.119, 0.219]	1.192	0.583	0.044
BRACS	Native	Severe (spread)	Difficulty	1.483	0.650	0.194 [0.148, 0.250]	1.199	0.589	0.041
BRACS	Native	Severe (spread)	Diversity	1.472	0.647	0.176 [0.128, 0.235]	1.208	0.592	0.041
BRACS	Native	Severe (spread)	Prevalence	1.000	0.491	0.067 [0.054, 0.100]	0.980	0.487	0.053
BRACS	—	Balanced	CE	0.993	0.486	0.051 [0.040, 0.087]	0.995	0.487	0.055
BRACS	—	Balanced	Nominal support	1.057	0.516	0.057 [0.043, 0.101]	1.040	0.511	0.045
BRACS	—	Balanced	Independent support	1.055	0.516	0.057 [0.043, 0.102]	1.038	0.511	0.045
BRACS	—	Balanced	Difficulty	1.057	0.516	0.058 [0.043, 0.105]	1.039	0.510	0.044
BRACS	—	Balanced	Diversity	1.045	0.511	0.048 [0.039, 0.087]	1.035	0.512	0.048
BRACS	—	Balanced	Prevalence	1.057	0.516	0.057 [0.042, 0.101]	1.040	0.511	0.045
BRACS	—	Natural	CE	0.925	0.443	0.069 [0.043, 0.107]	0.896	0.435	0.042

Table 31: Raw and temperature-scaled expected calibration error, negative log likelihood, and Brier score per condition and method, with crossed patient-bootstrap intervals on the calibration error. Temperature is fitted on natural-prevalence validation data and never gates.

Temperature scaling removes most aggregate miscalibration on the cross-entropy reference wherever the raw-score deficit meets the analysis criteria. Under difficulty-reversed severe, for example, it takes expected calibration error from 0.313 to 0.067 and negative log likelihood from 2.582 to 1.271. The balanced references change little: their calibration error is 0.040–0.051 before scaling, and scaling moves it by at most 0.006. This indicates that much of the observed probability-quality damage is a correctable confidence-scaling error induced by the allocation.

Table 31 reports aggregate metrics, however, and the gated endpoint is the tail group. The contrasts quoted in Section 3.3 therefore re-estimate the tail-group macro negative log likelihood itself on scaled probabilities, holding everything else fixed: the same confirmation runs, the same tail classes taken from each deprived condition's own allocation, the same frozen crossed patient bootstrap, and the same equal-weight average over the three partitions. Recomputing the raw contrasts by the same route reproduces the published values in Tables 18 and 25 exactly, which is what licenses reading the scaled column beside them. These scaled contrasts carry paired bootstrap intervals only. The partition-block permutation tests and the Holm adjustment were run on the prespecified raw endpoint and are not repeated here, so no scaled result is a confirmatory decision; the materiality minimum and the interval criterion are applied to them descriptively.

F Per-Partition Results

The main text averages the three partitions with equal weight, which is only defensible if no single partition drives a conclusion. This appendix reports the cross-entropy deficits separately for each partition, alongside the spread of achieved severity across them.

Dataset	Assignment	Severity	Gate	Split 0	Split 1	Split 2	Equal weight	ρ spread
BRACS	Native	Balanced	Discrimination	0.0252	0.0075	0.0547	0.0291	0.001
BRACS	Native	Balanced	Calibration	0.0702	0.1710	0.1714	0.1375	0.001
BRACS	Aligned	Balanced	Discrimination	0.0252	0.0067	0.0592	0.0303	0.001
BRACS	Aligned	Balanced	Calibration	0.0699	0.1483	0.2015	0.1399	0.001
BRACS	Reversed	Balanced	Discrimination	0.0200	0.0047	0.0539	0.0262	0.001
BRACS	Reversed	Balanced	Calibration	0.0597	0.1375	0.2042	0.1338	0.001
BRACS	Native	Moderate	Discrimination	0.0367	0.0353	0.0283	0.0334	0.000
BRACS	Native	Moderate	Calibration	0.0274	0.0921	0.4374	0.1856	0.000
BRACS	Aligned	Moderate	Discrimination	0.0163	0.0297	-0.0012	0.0150	0.000
BRACS	Aligned	Moderate	Calibration	0.5975	0.2429	0.2978	0.3794	0.000
BRACS	Reversed	Moderate	Discrimination	0.0861	0.0676	0.0914	0.0817	0.000
BRACS	Reversed	Moderate	Calibration	0.4098	0.0372	1.8136	0.7535	0.000
BRACS	Native	Severe	Discrimination	0.0467	0.0599	0.0569	0.0545	0.002
BRACS	Native	Severe	Calibration	0.4832	0.1602	1.4204	0.6879	0.002
BRACS	Aligned	Severe	Discrimination	0.0119	0.0730	0.0441	0.0430	0.002
BRACS	Aligned	Severe	Calibration	0.3062	0.9779	0.9042	0.7294	0.002
BRACS	Reversed	Severe	Discrimination	0.1288	0.1287	0.1394	0.1323	0.002
BRACS	Reversed	Severe	Calibration	1.2663	1.5668	2.3976	1.7436	0.002
BRACS	Native	Severe (spread)	Discrimination	0.0563	0.0998	0.0968	0.0843	0.002
BRACS	Native	Severe (spread)	Calibration	0.3566	0.1833	0.0747	0.2049	0.002
BRACS	Aligned	Severe (spread)	Discrimination	0.0556	0.0384	0.0304	0.0415	0.002
BRACS	Aligned	Severe (spread)	Calibration	1.1440	0.7027	0.8769	0.9078	0.002
BRACS	Reversed	Severe (spread)	Discrimination	0.1429	0.1160	0.1540	0.1377	0.002
BRACS	Reversed	Severe (spread)	Calibration	1.4890	0.9410	0.9559	1.1286	0.002

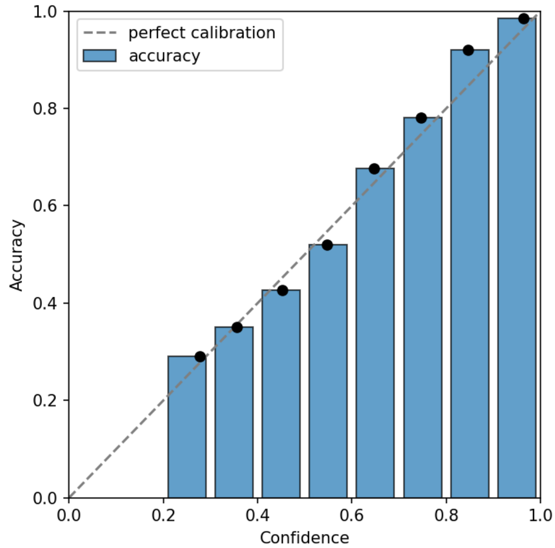
Table 33: Cross-entropy deficits by partition on both gate axes, with the equal-weight average and the relative achieved-severity spread across partitions, per condition and tail assignment.

No gate decision rests on a single partition, but the independent-support axis is the least uniform across them and this qualifies its reading. Its three units clear the 1.44-percentage-point discrimination threshold in two of the three partitions and fall to 0.30–0.64 percentage points on the second, against 5.40–5.97 on the third; the equal-weight average that opens the gate is therefore carried by two partitions rather than three. On the nominal axis the pattern is steadier: every gate-passing unit clears the threshold in all three partitions except difficulty-aligned severe, whose smallest partition-level deficit is 1.19 percentage points. On the calibration axis the three balanced units are consistent in falling below threshold, and among the eight gate-passing units the reversed moderate condition is the most variable, ranging from 0.037–1.814 nats. Achieved severity itself varies across partitions by at most 0.002 in relative terms, well inside the 0.5 consistency guard.

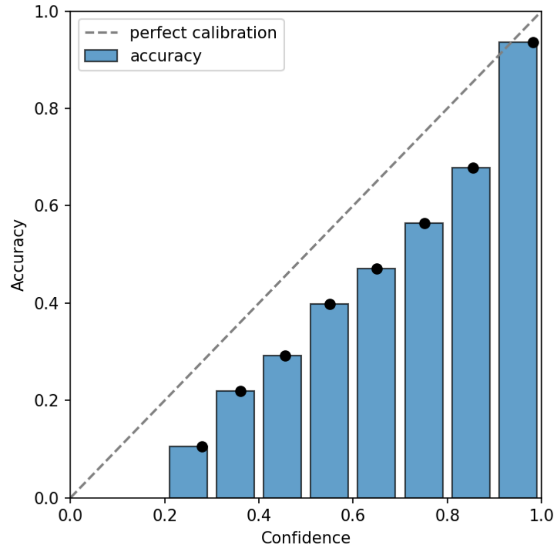
The conclusions of Section 3 are less exposed to this variation than the gate decisions are, because the recovery estimates on the independent-support axis are null in every partition rather than null only on average. The qualification attaches to the size of the independent-support deficit, not to the finding that nothing recovers it.

G Computational Cost

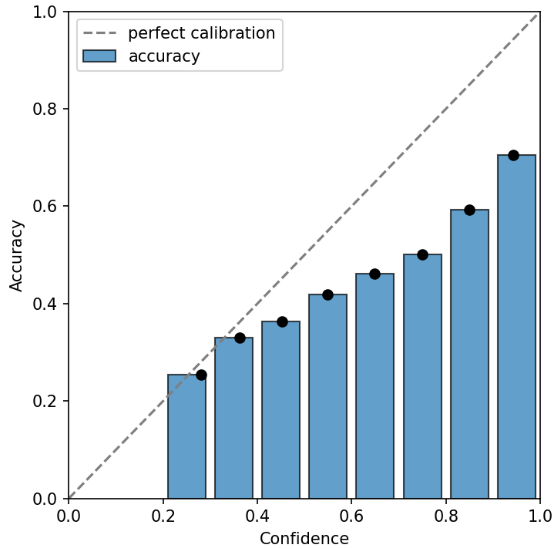
Cost bears on the design question because a method whose recovery requires substantially more computation than the reference is a different proposition from one that matches it. This appendix records processed examples, unique exposures, derived updates, examples per update, accelerator time, peak memory, and model size for every roster method, expressed as a matched difference against cross-entropy in the same condition.



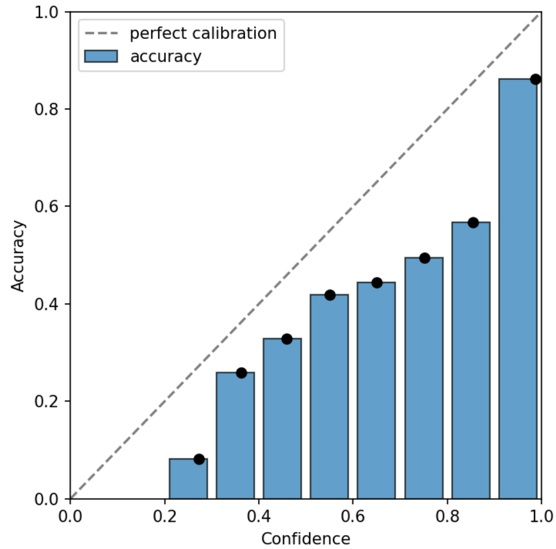
(a) Balanced reference, all classes



(b) Severe, native assignment, tail classes



(c) Severe, aligned assignment, tail classes



(d) Severe, reversed assignment, tail classes

Figure 1: Reliability curves for the balanced reference and for the tail classes under the severe condition of each tail assignment, with bin accuracy against mean predicted confidence over ten equal-width bins, on the first locked partition.

Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Balanced (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Balanced sampling	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Balanced (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Balanced sampling	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft F_1	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CE + soft MCC	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Aligned	Balanced (spread)	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Aligned	Balanced (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Aligned	Balanced (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Nominal support	Accelerator hours	0	[-0.001, 0]
BRACS	Aligned	Balanced (spread)	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Balanced (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Balanced (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Aligned	Balanced (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Aligned	Balanced (spread)	cRT	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Balanced (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Aligned	Balanced (spread)	OKO	Peak memory (bytes)	-1.731e+07	[-1.731e+07, -1.731e+07]
BRACS	Aligned	Balanced (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Aligned	Balanced (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Aligned	Balanced (spread)	OKO	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Balanced (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Aligned	Balanced (spread)	Diversity	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Diversity	Peak memory (bytes)	3.578e+07	[3.556e+07, 3.605e+07]
BRACS	Aligned	Balanced (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Balanced (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Balanced (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Balanced sampling	Unique exposures	-0.067	[-0.2, 0]
BRACS	Aligned	Moderate	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Aligned	Moderate	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	CE + soft F_1	Unique exposures	-0.067	[-0.2, 0]
BRACS	Aligned	Moderate	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.002]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Moderate	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	CE + soft MCC	Unique exposures	-0.067	[-0.2, 0]
BRACS	Aligned	Moderate	CFAL	Accelerator hours	0.004	[0.004, 0.005]
BRACS	Aligned	Moderate	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Aligned	Moderate	CFAL	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	CFAL	Parameters	-7	[-7, -7]
BRACS	Aligned	Moderate	CFAL	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Moderate	cRT	Peak memory (bytes)	-2.019e+07	[-2.019e+07, -2.019e+07]
BRACS	Aligned	Moderate	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Aligned	Moderate	cRT	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	cRT	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Moderate	Focal loss	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	Focal loss	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Focal loss	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Focal loss	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Aligned	Moderate	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Aligned	Moderate	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (train)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	OKO	Accelerator hours	0.001	[0, 0.001]
BRACS	Aligned	Moderate	OKO	Peak memory (bytes)	-3.54e+05	[-3.54e+05, -3.54e+05]
BRACS	Aligned	Moderate	OKO	Processed examples	2983	[2983, 2983]
BRACS	Aligned	Moderate	OKO	Parameters	3591	[3591, 3591]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Moderate	OKO	Unique exposures	-0.067	[-0.2, 0]
BRACS	Aligned	Moderate	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Aligned	Moderate	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Aligned	Moderate	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Moderate	Diversity	Peak memory (bytes)	-1.147e+07	[-1.152e+07, -1.141e+07]
BRACS	Aligned	Moderate	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Moderate	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Moderate	Prevalence	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe	CE + soft F_1	Accelerator hours	0.005	[0.004, 0.005]
BRACS	Aligned	Severe	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Aligned	Severe	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Severe	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Aligned	Severe	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe	CFAL	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Aligned	Severe	CFAL	Peak memory (bytes)	1.138e+09	[1.138e+09, 1.138e+09]
BRACS	Aligned	Severe	CFAL	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	CFAL	Parameters	-7	[-7, -7]
BRACS	Aligned	Severe	CFAL	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Peak memory (bytes)	512	[512, 512]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Severe	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Aligned	Severe	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Aligned	Severe	cRT	Parameters	0	[0, 0]
BRACS	Aligned	Severe	cRT	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Focal loss	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Severe	Focal loss	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Focal loss	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Focal loss	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	OKO	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	OKO	Peak memory (bytes)	3.261e+07	[3.261e+07, 3.261e+07]
BRACS	Aligned	Severe	OKO	Processed examples	2983	[2983, 2983]
BRACS	Aligned	Severe	OKO	Parameters	3591	[3591, 3591]
BRACS	Aligned	Severe	OKO	Unique exposures	-8.933	[-10.47, -7.467]
BRACS	Aligned	Severe	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Severe	Diversity	Accelerator hours	0	[0, 0.001]
BRACS	Aligned	Severe	Diversity	Peak memory (bytes)	8.334e+05	[8.238e+05, 8.43e+05]
BRACS	Aligned	Severe	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Aligned	Severe	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Severe	Prevalence	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Aligned	Severe (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Aligned	Severe (spread)	CFAL	Peak memory (bytes)	1.105e+09	[1.105e+09, 1.105e+09]
BRACS	Aligned	Severe (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Aligned	Severe (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Nominal support	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Aligned	Severe (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	cRT	Accelerator hours	0.001	[0.001, 0.002]
BRACS	Aligned	Severe (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Aligned	Severe (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Aligned	Severe (spread)	cRT	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	Focal loss	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Aligned	Severe (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Aligned	Severe (spread)	OKO	Peak memory (bytes)	-9.001e+05	[-9.001e+05, -9.001e+05]
BRACS	Aligned	Severe (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Aligned	Severe (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Aligned	Severe (spread)	OKO	Unique exposures	-26.4	[-28.47, -24.13]
BRACS	Aligned	Severe (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Difficulty	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Aligned	Severe (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Aligned	Severe (spread)	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Aligned	Severe (spread)	Diversity	Peak memory (bytes)	1.155e+06	[1.07e+06, 1.241e+06]
BRACS	Aligned	Severe (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Aligned	Severe (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Aligned	Severe (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Aligned	Severe (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Aligned	Severe (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Aligned	Severe (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Balanced sampling	Accelerator hours	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Balanced (spread)	Balanced sampling	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Balanced (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Balanced sampling	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft F_1	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CE + soft MCC	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Reversed	Balanced (spread)	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Reversed	Balanced (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Reversed	Balanced (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Balanced (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Balanced (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Reversed	Balanced (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Reversed	Balanced (spread)	cRT	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Balanced (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Parameters	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Balanced (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Reversed	Balanced (spread)	OKO	Peak memory (bytes)	-1.731e+07	[-1.731e+07, -1.731e+07]
BRACS	Reversed	Balanced (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Reversed	Balanced (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Reversed	Balanced (spread)	OKO	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Balanced (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Reversed	Balanced (spread)	Diversity	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Diversity	Peak memory (bytes)	3.579e+07	[3.556e+07, 3.605e+07]
BRACS	Reversed	Balanced (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Balanced (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Balanced (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Balanced sampling	Unique exposures	-0.067	[-0.2, 0]
BRACS	Reversed	Moderate	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Moderate	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	CE + soft F_1	Unique exposures	-0.067	[-0.2, 0]
BRACS	Reversed	Moderate	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Moderate	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Moderate	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	CE + soft MCC	Unique exposures	-0.067	[-0.2, 0]
BRACS	Reversed	Moderate	CFAL	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Moderate	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Reversed	Moderate	CFAL	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	CFAL	Parameters	-7	[-7, -7]
BRACS	Reversed	Moderate	CFAL	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	Nominal support	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Moderate	cRT	Peak memory (bytes)	-2.019e+07	[-2.019e+07, -2.019e+07]
BRACS	Reversed	Moderate	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Reversed	Moderate	cRT	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	cRT	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Focal loss	Accelerator hours	0	[0, 0.001]
BRACS	Reversed	Moderate	Focal loss	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	Focal loss	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Focal loss	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Focal loss	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Reversed	Moderate	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Reversed	Moderate	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (train)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	OKO	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	OKO	Peak memory (bytes)	-3.54e+05	[-3.54e+05, -3.54e+05]
BRACS	Reversed	Moderate	OKO	Processed examples	2983	[2983, 2983]
BRACS	Reversed	Moderate	OKO	Parameters	3591	[3591, 3591]
BRACS	Reversed	Moderate	OKO	Unique exposures	-0.067	[-0.2, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Moderate	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Reversed	Moderate	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Reversed	Moderate	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Moderate	Diversity	Peak memory (bytes)	-9.537e+06	[-9.563e+06, -9.516e+06]
BRACS	Reversed	Moderate	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Moderate	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Moderate	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Severe	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Reversed	Severe	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Severe	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Reversed	Severe	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe	CFAL	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Reversed	Severe	CFAL	Peak memory (bytes)	1.138e+09	[1.138e+09, 1.138e+09]
BRACS	Reversed	Severe	CFAL	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	CFAL	Parameters	-7	[-7, -7]
BRACS	Reversed	Severe	CFAL	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe	Nominal support	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Severe	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Reversed	Severe	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Reversed	Severe	cRT	Parameters	0	[0, 0]
BRACS	Reversed	Severe	cRT	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Focal loss	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Severe	Focal loss	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Focal loss	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Focal loss	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	OKO	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	OKO	Peak memory (bytes)	3.261e+07	[3.261e+07, 3.261e+07]
BRACS	Reversed	Severe	OKO	Processed examples	2983	[2983, 2983]
BRACS	Reversed	Severe	OKO	Parameters	3591	[3591, 3591]
BRACS	Reversed	Severe	OKO	Unique exposures	-16.2	[-17.8, -14.67]
BRACS	Reversed	Severe	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Reversed	Severe	Diversity	Accelerator hours	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Severe	Diversity	Peak memory (bytes)	1.152e+06	[1.15e+06, 1.153e+06]
BRACS	Reversed	Severe	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Reversed	Severe	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Severe	Prevalence	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Reversed	Severe (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Reversed	Severe (spread)	CFAL	Peak memory (bytes)	1.105e+09	[1.105e+09, 1.105e+09]
BRACS	Reversed	Severe (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Reversed	Severe (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Nominal support	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Reversed	Severe (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	cRT	Accelerator hours	0.001	[0.001, 0.002]
BRACS	Reversed	Severe (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Reversed	Severe (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Reversed	Severe (spread)	cRT	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Focal loss	Parameters	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Reversed	Severe (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Reversed	Severe (spread)	OKO	Peak memory (bytes)	-9.001e+05	[-9.001e+05, -9.001e+05]
BRACS	Reversed	Severe (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Reversed	Severe (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Reversed	Severe (spread)	OKO	Unique exposures	-79.53	[-84.07, -74.87]
BRACS	Reversed	Severe (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Difficulty	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Reversed	Severe (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Reversed	Severe (spread)	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Reversed	Severe (spread)	Diversity	Peak memory (bytes)	1.129e+06	[1.119e+06, 1.137e+06]
BRACS	Reversed	Severe (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Reversed	Severe (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Reversed	Severe (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Reversed	Severe (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Reversed	Severe (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Reversed	Severe (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Balanced sampling	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Balanced (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Balanced sampling	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Balanced (spread)	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Balanced (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft F_1	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Balanced (spread)	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Balanced (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	CE + soft MCC	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Native	Balanced (spread)	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Native	Balanced (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Native	Balanced (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Balanced (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Balanced (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Native	Balanced (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Native	Balanced (spread)	cRT	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Focal loss	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Balanced (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Native	Balanced (spread)	OKO	Peak memory (bytes)	-1.731e+07	[-1.731e+07, -1.731e+07]
BRACS	Native	Balanced (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Native	Balanced (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Native	Balanced (spread)	OKO	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Balanced (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Native	Balanced (spread)	Diversity	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Diversity	Peak memory (bytes)	3.579e+07	[3.556e+07, 3.605e+07]
BRACS	Native	Balanced (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Balanced (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Balanced (spread)	Prevalence	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Balanced (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Balanced (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Balanced (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Parameters	0	[0, 0]
BRACS	Native	Moderate	Balanced sampling	Unique exposures	-0.067	[-0.2, 0]
BRACS	Native	Moderate	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Moderate	CE + soft F_1	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Native	Moderate	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Native	Moderate	CE + soft F_1	Unique exposures	-0.067	[-0.2, 0]
BRACS	Native	Moderate	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Moderate	CE + soft MCC	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	CE + soft MCC	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Moderate	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Native	Moderate	CE + soft MCC	Unique exposures	-0.067	[-0.2, 0]
BRACS	Native	Moderate	CFAL	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Moderate	CFAL	Peak memory (bytes)	1.088e+09	[1.088e+09, 1.088e+09]
BRACS	Native	Moderate	CFAL	Processed examples	0	[0, 0]
BRACS	Native	Moderate	CFAL	Parameters	-7	[-7, -7]
BRACS	Native	Moderate	CFAL	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Parameters	0	[0, 0]
BRACS	Native	Moderate	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Moderate	cRT	Peak memory (bytes)	-2.019e+07	[-2.019e+07, -2.019e+07]
BRACS	Native	Moderate	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Native	Moderate	cRT	Parameters	0	[0, 0]
BRACS	Native	Moderate	cRT	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Focal loss	Accelerator hours	0.001	[0, 0.001]
BRACS	Native	Moderate	Focal loss	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	Focal loss	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Focal loss	Parameters	0	[0, 0]
BRACS	Native	Moderate	Focal loss	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Independent support	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Native	Moderate	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Independent support	Parameters	0	[0, 0]
BRACS	Native	Moderate	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Independent support (matched β)	Peak memory (bytes)	-2.179e+05	[-2.179e+05, -2.179e+05]
BRACS	Native	Moderate	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Native	Moderate	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (train)	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	OKO	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	OKO	Peak memory (bytes)	-3.54e+05	[-3.54e+05, -3.54e+05]
BRACS	Native	Moderate	OKO	Processed examples	2983	[2983, 2983]
BRACS	Native	Moderate	OKO	Parameters	3591	[3591, 3591]
BRACS	Native	Moderate	OKO	Unique exposures	-0.067	[-0.2, 0]
BRACS	Native	Moderate	Difficulty	Accelerator hours	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Moderate	Difficulty	Peak memory (bytes)	-1.669e+07	[-1.669e+07, -1.669e+07]
BRACS	Native	Moderate	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Moderate	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.493e+09	[-3.493e+09, -3.493e+09]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Native	Moderate	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Native	Moderate	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Moderate	Diversity	Peak memory (bytes)	-7.995e+06	[-7.997e+06, -7.991e+06]
BRACS	Native	Moderate	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Diversity	Parameters	0	[0, 0]
BRACS	Native	Moderate	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Moderate	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Moderate	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Parameters	0	[0, 0]
BRACS	Native	Severe	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Severe	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Native	Severe	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Native	Severe	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe	CE + soft MCC	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Severe	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Native	Severe	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Native	Severe	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe	CFAL	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Native	Severe	CFAL	Peak memory (bytes)	1.138e+09	[1.138e+09, 1.138e+09]
BRACS	Native	Severe	CFAL	Processed examples	0	[0, 0]
BRACS	Native	Severe	CFAL	Parameters	-7	[-7, -7]
BRACS	Native	Severe	CFAL	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Nominal support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Severe	Nominal support	Parameters	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Severe	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Severe	cRT	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Native	Severe	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Native	Severe	cRT	Parameters	0	[0, 0]
BRACS	Native	Severe	cRT	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Focal loss	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Focal loss	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Severe	Focal loss	Processed examples	0	[0, 0]
BRACS	Native	Severe	Focal loss	Parameters	0	[0, 0]
BRACS	Native	Severe	Focal loss	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Severe	Independent support	Parameters	0	[0, 0]
BRACS	Native	Severe	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Native	Severe	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Native	Severe	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Native	Severe	OKO	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	OKO	Peak memory (bytes)	3.261e+07	[3.261e+07, 3.261e+07]
BRACS	Native	Severe	OKO	Processed examples	2983	[2983, 2983]
BRACS	Native	Severe	OKO	Parameters	3591	[3591, 3591]
BRACS	Native	Severe	OKO	Unique exposures	-14.47	[-16.27, -12.6]
BRACS	Native	Severe	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Difficulty	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Severe	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Severe	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Native	Severe	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Native	Severe	Diversity	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Diversity	Peak memory (bytes)	1.152e+06	[1.15e+06, 1.153e+06]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Severe	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Severe	Diversity	Parameters	0	[0, 0]
BRACS	Native	Severe	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Severe	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Severe	Prevalence	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	Native	Severe	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Severe	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Severe	Prevalence	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Balanced sampling	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe (spread)	CE + soft F_1	Accelerator hours	0.004	[0.004, 0.004]
BRACS	Native	Severe (spread)	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft F_1	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft F_1	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe (spread)	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe (spread)	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft MCC	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	CE + soft MCC	Unique exposures	-3.733	[-4.6, -2.933]
BRACS	Native	Severe (spread)	CFAL	Accelerator hours	0.003	[0.003, 0.003]
BRACS	Native	Severe (spread)	CFAL	Peak memory (bytes)	1.105e+09	[1.105e+09, 1.105e+09]
BRACS	Native	Severe (spread)	CFAL	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	CFAL	Parameters	-7	[-7, -7]
BRACS	Native	Severe (spread)	CFAL	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Nominal support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Nominal support	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Native	Severe (spread)	Nominal support	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Nominal support	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Nominal support	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	cRT	Accelerator hours	0.001	[0.001, 0.002]
BRACS	Native	Severe (spread)	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	Native	Severe (spread)	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	Native	Severe (spread)	cRT	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	cRT	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe (spread)	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	Focal loss	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Focal loss	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Focal loss	Unique exposures	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	Native	Severe (spread)	Independent support	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe (spread)	Independent support	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe (spread)	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	Native	Severe (spread)	OKO	Peak memory (bytes)	-9.001e+05	[-9.001e+05, -9.001e+05]
BRACS	Native	Severe (spread)	OKO	Processed examples	2727	[2727, 2727]
BRACS	Native	Severe (spread)	OKO	Parameters	3591	[3591, 3591]
BRACS	Native	Severe (spread)	OKO	Unique exposures	-70.87	[-75.87, -65.8]
BRACS	Native	Severe (spread)	Difficulty	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Difficulty	Peak memory (bytes)	1.647e+07	[1.647e+07, 1.647e+07]
BRACS	Native	Severe (spread)	Difficulty	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Difficulty	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Difficulty	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	Native	Severe (spread)	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	Native	Severe (spread)	Diversity	Peak memory (bytes)	1.129e+06	[1.119e+06, 1.137e+06]
BRACS	Native	Severe (spread)	Diversity	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Diversity	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Diversity	Unique exposures	0	[0, 0]
BRACS	Native	Severe (spread)	Prevalence	Accelerator hours	0	[0, 0]
BRACS	Native	Severe (spread)	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	Native	Severe (spread)	Prevalence	Processed examples	0	[0, 0]
BRACS	Native	Severe (spread)	Prevalence	Parameters	0	[0, 0]
BRACS	Native	Severe (spread)	Prevalence	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Balanced sampling	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Balanced sampling	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	Balanced sampling	Processed examples	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	—	Balanced	Balanced sampling	Parameters	0	[0, 0]
BRACS	—	Balanced	Balanced sampling	Unique exposures	0	[0, 0]
BRACS	—	Balanced	CE + soft F_1	Accelerator hours	0.014	[0.014, 0.014]
BRACS	—	Balanced	CE + soft F_1	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	CE + soft F_1	Processed examples	0	[0, 0]
BRACS	—	Balanced	CE + soft F_1	Parameters	0	[0, 0]
BRACS	—	Balanced	CE + soft F_1	Unique exposures	0	[0, 0]
BRACS	—	Balanced	CE + soft MCC	Accelerator hours	0.001	[0.001, 0.002]
BRACS	—	Balanced	CE + soft MCC	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	CE + soft MCC	Processed examples	0	[0, 0]
BRACS	—	Balanced	CE + soft MCC	Parameters	0	[0, 0]
BRACS	—	Balanced	CE + soft MCC	Unique exposures	0	[0, 0]
BRACS	—	Balanced	CFAL	Accelerator hours	0.004	[0.004, 0.004]
BRACS	—	Balanced	CFAL	Peak memory (bytes)	1.121e+09	[1.121e+09, 1.121e+09]
BRACS	—	Balanced	CFAL	Processed examples	0	[0, 0]
BRACS	—	Balanced	CFAL	Parameters	-7	[-7, -7]
BRACS	—	Balanced	CFAL	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Nominal support	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Nominal support	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	—	Balanced	Nominal support	Processed examples	0	[0, 0]
BRACS	—	Balanced	Nominal support	Parameters	0	[0, 0]
BRACS	—	Balanced	Nominal support	Unique exposures	0	[0, 0]
BRACS	—	Balanced	cRT	Accelerator hours	0.002	[0.002, 0.002]
BRACS	—	Balanced	cRT	Peak memory (bytes)	-3.497e+06	[-3.497e+06, -3.497e+06]
BRACS	—	Balanced	cRT	Processed examples	129.3	[129.3, 129.3]
BRACS	—	Balanced	cRT	Parameters	0	[0, 0]
BRACS	—	Balanced	cRT	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Focal loss	Accelerator hours	0.001	[0.001, 0.001]
BRACS	—	Balanced	Focal loss	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	Focal loss	Processed examples	0	[0, 0]
BRACS	—	Balanced	Focal loss	Parameters	0	[0, 0]
BRACS	—	Balanced	Focal loss	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Independent support	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Independent support	Peak memory (bytes)	512	[512, 512]
BRACS	—	Balanced	Independent support	Processed examples	0	[0, 0]
BRACS	—	Balanced	Independent support	Parameters	0	[0, 0]
BRACS	—	Balanced	Independent support	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Independent support (matched β)	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Independent support (matched β)	Peak memory (bytes)	512	[512, 512]
BRACS	—	Balanced	Independent support (matched β)	Processed examples	0	[0, 0]
BRACS	—	Balanced	Independent support (matched β)	Parameters	0	[0, 0]
BRACS	—	Balanced	Independent support (matched β)	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (train)	Accelerator hours	0	[0, 0]

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Dataset	Assignment	Condition	Method	Endpoint	Effect	95% CI
BRACS	—	Balanced	Logit adjustment (train)	Peak memory (bytes)	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (train)	Processed examples	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (train)	Parameters	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (train)	Unique exposures	0	[0, 0]
BRACS	—	Balanced	OKO	Accelerator hours	0.002	[0.002, 0.002]
BRACS	—	Balanced	OKO	Peak memory (bytes)	-6.161e+05	[-6.161e+05, -6.161e+05]
BRACS	—	Balanced	OKO	Processed examples	2727	[2727, 2727]
BRACS	—	Balanced	OKO	Parameters	3591	[3591, 3591]
BRACS	—	Balanced	OKO	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Difficulty	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Difficulty	Peak memory (bytes)	1.669e+07	[1.669e+07, 1.669e+07]
BRACS	—	Balanced	Difficulty	Processed examples	0	[0, 0]
BRACS	—	Balanced	Difficulty	Parameters	0	[0, 0]
BRACS	—	Balanced	Difficulty	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Accelerator hours	-0.003	[-0.003, -0.003]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Peak memory (bytes)	-3.476e+09	[-3.476e+09, -3.476e+09]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Processed examples	-5.774e+05	[-5.774e+05, -5.774e+05]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Parameters	0	[0, 0]
BRACS	—	Balanced	Logit adjustment (post-hoc)	Unique exposures	-1.933e+04	[-1.933e+04, -1.933e+04]
BRACS	—	Balanced	Diversity	Accelerator hours	0.001	[0.001, 0.001]
BRACS	—	Balanced	Diversity	Peak memory (bytes)	1.817e+07	[1.744e+07, 1.883e+07]
BRACS	—	Balanced	Diversity	Processed examples	0	[0, 0]
BRACS	—	Balanced	Diversity	Parameters	0	[0, 0]
BRACS	—	Balanced	Diversity	Unique exposures	0	[0, 0]
BRACS	—	Balanced	Prevalence	Accelerator hours	0	[0, 0]
BRACS	—	Balanced	Prevalence	Peak memory (bytes)	512	[512, 512]
BRACS	—	Balanced	Prevalence	Processed examples	0	[0, 0]
BRACS	—	Balanced	Prevalence	Parameters	0	[0, 0]
BRACS	—	Balanced	Prevalence	Unique exposures	0	[0, 0]

Table 35: Matched computational-cost comparisons against cross-entropy in the same condition, for every roster method and every recorded cost endpoint, with crossed patient-bootstrap intervals.

The exposure budget is matched to within 0.6% across the roster, which is the property that makes the recovery comparison a comparison of methods rather than of data volume. Every method processes the same 503,880 examples as cross-entropy to within rounding, except set learning, which exceeds it by 2,727 to 2,983 examples because its per-update example count depends on a tuned set size and the derived update count must be integral, and classifier retraining, which exceeds it by 129. Post-hoc logit adjustment trains nothing of its own and reuses the cross-entropy checkpoint, so its processed-example difference is the full budget with a negative sign; its cost is a single pass to fit one scalar.

That matched exposure is achieved through unequal update counts, which is the intended trade. Set learning takes roughly 3,800 fewer updates than cross-entropy at 641 more examples per update, and classifier retraining splits its budget between two stages in a fixed ratio. Accelerator time, peak memory, and parameter counts differ by amounts their intervals absorb in every case, so no roster method's recovery on this dataset is bought with materially more computation than the reference.

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